

A perturbation proteomics-based foundation model for virtual cell construction

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Abstract

Building a virtual cell requires comprehensive understanding of protein network dynamics of a cell which necessitates large-scale perturbation proteome data and intelligent computational models learned from the proteome data corpus. Here, we generate a large-scale dataset of over 38 million perturbed protein measurements in breast cancer cell lines and develop a neural ordinary differential equation-based foundation model, namely ProteinTalks. During pretraining, ProteinTalks gains a fundamental understanding of cellular protein network dynamics. Our model encodes protein networks and exhibits consistently improved predictive accuracy across various downstream tasks, highlighting its generalization capabilities and adaptability. In cancer cells, ProteinTalks robustly predicts drug efficacy and synergy, identifies novel drug combinations, and, through its interpretability, uncovers resistance-associated proteins. When applied to more complex system, patient-derived tumor xenografts, ProteinTalks predicts potential responses to drugs. Its integration with clinical patient data enhances the prognosis prediction of breast cancer patients. Collectively, we present a foundational model based on proteome dynamics, offering potential for various

downstream applications, including drug discovery, and providing a basis for developing virtual cells.

Introduction

Virtual cell refers to a computational model of a physical cell, designed to simulate and predict cellular processes *in silico*¹⁻³. Constructing a virtual cell requires thorough time-resolved measurements of all cell ingredients, reflecting the cellular functions and dynamic behaviors⁴. Protein networks are essential for understanding cell biology and disease mechanisms, providing insights into disease progression and informing treatment strategies^{5,6}. However, large-scale proteomic data, particularly including dynamic information, remains extremely sparse compared to transcriptomic data. In drug discovery, targeting specific proteins without considering their network context may result in limited efficacy or unintended side effects⁷. Consequently, systematic analysis and modeling of proteomic dynamics are crucial for identifying novel drug targets, designing more effective and precise therapies, and ultimately developing comprehensive virtual cell models.

Perturbation proteomics provides a powerful approach to decipher complex protein network dynamics, aiding drug discovery by uncovering drug mechanisms of action (MOAs)⁸⁻¹². With advances in high-throughput proteomics¹³ enabled by data-independent acquisition mass spectrometry (DIA-MS)¹⁴, generating large-scale perturbation proteomics datasets is now achievable⁹, paved way for utilizing large scale pre-training techniques to describe the protein dynamical space.

Here, we integrated a multidimensional perturbation approach to comprehend the complexity of interconnected drug-protein systems. Utilizing a 96-well-based high-throughput platform, we perturbed breast cancer cell lines with clinically relevant drugs and their combinations, and subsequently obtained over 38 million perturbed protein measurements, as well as cell morphological and viability data. Based on this large-scale proteome data corpus, we further developed a dynamical foundation model called ProteinTalks, based on ordinary differential equation (ODE) network models¹⁵, to predict perturbed proteomic networks, then extend to identification of essential proteins associated with drug efficacy, and characterization of cellular response after drug treatment. Since the dynamical information was explicitly integrated in the model architecture, our model demonstrated excellent generalizability and adaptability in multiple downstream applications in cell lines, mouse models and patients, with significantly fewer parameters. Overall, this study introduces a paradigm shift to combine large scale perturbation data and dynamical neural network to expand the concept of foundation models beyond scale to include the dimension of time. We believe such rationale will serve as an important alternative to effectively make use of limited measurements to build limitless virtual cell models across space and time.

Results

Generation of perturbation proteome datasets for building a foundation model

As a proof of concept, and to be directly related to drug development, we focused specifically

on cancer cell lines, specifically breast cancer cells (**Figure 1A**). We selected 16 commonly used TNBC cell lines and two non-TNBC cell lines (**Table S1A**). We also collected 63 FDA-approved small-molecule drugs commonly used in the clinical treatment of breast cancer, which were categorized into three primary classes and 20 subclasses, as detailed in **Table S1B**, and involved in various signaling pathways (**Figure S1A**). Based on a previous drug combination screening study¹⁶, we selected 914 combination-cell line tuples across 18 common breast cancer cell lines. Each drug pair included one drug at two different concentrations to achieve 50-90% cell viability for each cell line, alongside a second drug at a single concentration¹⁶. Additionally, we incorporated 98 anti-cancer compounds (**Table S1B**), which were either FDA-approved, in clinical trials, or under investigation for breast cancer, pancreatic cancer, colon cancer, and lung cancer¹⁶.

We treated the 18 cell lines with 63 FDA-approved drugs at 6, 24, and 48 hours, each in triplicates (**Figure 1A**). Approximately 10% of randomly selected samples were injected in replicates for quality control. Multiple control samples were analyzed to assess and correct batch effects arising from sample preparation and DIA-MS analysis. Altogether, we acquired 16,311 perturbation proteomic DIA-MS data files with approximately 6668 hours of mass spectrometry instrument time, yielding over 38 million high-quality perturbed protein measurements (**Figure 1A**). With DIA-NN analyses, the data led to the relative quantification of 5530 protein groups, corresponding to 5143 unique proteins (**Table S2A**). Data quality analyses are presented in **Figure S1**. Our proteomics dataset includes 252 proteins with known specific relevance to TNBC biology when compared to non-TNBC cells (**Table S2B**), 100 proteins characteristic in the basal A subtype, and 14 proteins specific to the basal B subtype (**Table S2C**), based on the seminal study conducted by Neel and colleagues¹⁷. In addition, we collected 23,482 data points from cytotoxicity assays to complement the proteomic analysis. The quality control analysis demonstrated consistent reproducibility, validating the inclusion of protein measurements from biological replicates (**Figures S1B-C**). Our large-scale proteomic analysis revealed distinct differences among cell lines (**Figure 1B**), while variations in drug treatment durations (**Figure 1C**) and drug interference (**Figures 1D, S1E**) showed minimal impact on proteomic profiles. The resulting proteome datasets, termed ProteinTalks datasets (PTDS), are available at db.prottalks.com.

The perturbation proteomics datasets capture drug-induced modulation of protein networks

To discern whether perturbed proteomes might signify the MOA for drugs, we firstly focused on the proteins directly targeted by the compounds. We identified 61 protein targets, accounting for 48.8% of the targets of the 63 drugs under investigation (**Figure S2A**). Intriguingly, in cells resistant to capecitabine, thymidylate synthetase (TYMS) – a known target of the drug – demonstrated a substantial increase throughout the treatment duration (**Figure S2B**). To verify its role in drug resistance, we knocked down TYMS in the HCC1143 cell line and observed enhanced sensitivity to capecitabine (**Figures S2B-C**). The data imply that our perturbation proteomics data can illuminate potential molecular mechanisms responsible for drug efficacy and resistance.

Beyond the proteins directly targeted by the drugs, we also detected broader proteomic changes, indicating the indirect effects of drug treatment. We developed a quantitative metric

called PertScore that encapsulates the aggregate prevalence of protein changes elicited by various perturbation conditions, encompassing drug types, treatment durations, and cell lines (see **Methods**). Using PertScore, we could pinpoint the most recurrent protein alterations stemming from drug perturbations (**Figure S2D**). We identified a total of 1123 perturbation-related proteins (PertScore > 10) (**Figures S2E, Table S3**). Upon enriching pathways, we observed that MOA-associated pathways were significantly perturbed at 6 hours, while cell death-related pathways were enriched at 48 hours (**Figure 2A**). Our data suggest that proteins in certain pathways act as vanguards for phenotypic shifts in cancer cells. For example, proteins differentially expressed due to alkylation agents are predominantly enriched in the DNA damage repair pathway, while hormone-based treatments result in changes primarily in the lipid metabolism pathway. Proteins affected by microtubule inhibitors show enrichment in the cellular cytoskeleton and associated pathways. Moreover, proteins with altered expression resulting from CDK inhibitor treatment are enriched in cell cycle-related pathways.

Next, we narrowed down to the consistently dysregulated proteins resulting from the drug subtype perturbations through mFuzz analysis. Our findings indicate that both chemotherapy and targeted drugs increase the expression levels of proteins related to nucleic acid synthesis and fatty acid metabolism, while decreasing the expression levels of proteins involved in RNA processing, RNA metabolism, and cell cycle progression (**Figure 2B**). The ascending proteins along the treatment of kinase inhibitors are linked to various kinase-associated signaling pathways such as MET, NOTCH4, and mTOR. Diverse cell death pathways are also upregulated with drug treatment (**Figure 2B**). Therefore, each compound interacts with one or more protein targets and induces proteome-level changes via both common and perturbagen-specific modulation of cellular processes that reflect its MOA. These perturbagen-induced abundance changes vary in magnitude from protein to protein.

Proteome dynamics is associated with drug resistance

Furthermore, we explored longitudinal changes related to drug sensitivity within this dataset. A total of 113 proteins exhibited differentially dynamic expression in response to drug treatment, effectively distinguishing between the chemotherapy-resistant and sensitive groups (**Table S3**). Specifically, 75 proteins exhibited an increase in expression following perturbation in the resistant group but decreased in the sensitive group (**Figure 2C**). These proteins were mainly enriched in cell cycle associated pathways, protein localization, purine ribonucleotide metabolic process, and carbon metabolism (**Figure 2C**). Conversely, 38 proteins showed an inverse dynamic expression pattern and were mainly associated with small molecule transport, endomembrane system organization, and the AKT signaling pathway (**Figure 2C**).

Multiple proteins maintained high expression levels in the resistant group while decreased in the sensitive group (**Figure 2D**), suggesting their involvement in drug resistance. For example, *ATG3*, an autophagy-related gene, showed a different dynamic trend between the two groups in response to 5-Fluorouracil, an anti-metabolic agent (**Figure 2D**). *ATG3* has been implicated in chemotherapy resistance in HCC treatment by regulating autophagy processes¹⁸. *BTF3* also exhibited a distinct pattern in expression between two groups treated with Docetaxel, an anti-mitotic agent (**Figure 2D**). The upregulation of *BTF3* indicates that the epithelial cancer

cells may possess stemness characteristics¹⁹, while cancer stem cells have shown chemotherapy resistance²⁰. PAK1 and NDE1 manifested this different dynamic trend after perturbation of Erlotinib, an EGFR inhibitor (**Figure 2D**). PAK1 has been associated with resistance to tyrosine kinase inhibitors in EGFR mutant lung cancer²¹. NDE1 has been verified to interact with EGFR²², indicating that the expression of NDE1 might influence the efficacy of the EGFR inhibitor Erlotinib (**Figure 2D**). ELAVL2 responded differently in the two groups when treated with Talazoparib, a PARP inhibitor (**Figure 2D**).

ELAVL2 has been implicated in drug resistance through the regulation of glycolysis²³. Conversely, the expression of CKS2 was elevated in the sensitive group while decreased in the resistant group (**Figure 2D**), highlighting its opposite effect on drug resistant compared with the previously mentioned proteins. Consistent with the published study, CKS2 exerts an antagonistic effect on the PI3K/Akt pathways²⁴.

178 **Development of a dynamical foundation model ProteinTalks**

After benchmarking the proteomics dataset, we then established a dynamical foundation model, namely ProteinTalks, to achieve systematic understanding of protein network dynamics using an ordinary differential equation (ODE) integrated with the perturbation-aware neural network. We pretrained ProteinTalks using a dataset of 38 million protein measurements from 16,311 perturbed proteomic data and the characteristics of the drugs. The model contains two modules (**Figure 3A**). The first module incorporates baseline proteome data from untreated cell lines and drug targets. Through an encoder, the model is trained to predict perturbed proteomes at multiple time points. These predictions are then compared to the ground-truth proteome data, resulting in the calculation of a mean squared error (MSE) loss, referred to as Loss₁ (**Figure S3**). This module stores the information of the protein network dynamics. In the second module, the predicted perturbed proteomes, combined with the structure of these drugs, including 881-dimensional drug molecular fingerprints (DMF), 55-dimensional drug physicochemical properties (DDP) and 61 targets of 63 drugs, are used to learn the cellular response to various perturbagens and the core proteins associated with drug response through a multilayer perception (MLP) (**Figure 3A, S3**). More technical details of the ProteinTalks model construction are provided in **Methods**. This ProteinTalks foundation model enables a wide range of applications in drug discovery and precision medicine.

197 **Prediction of responsiveness of drugs**

We tested whether ProteinTalks model could boost the prediction of drug responsiveness through protein network dynamics in the biological systems based on unperturbed omics data. Notably, the ProteinTalks outperformed typical machine learning models, including bootstrap, random forest, logistic regression, SGD, KNN, and DeepSynergy (area under the receiver operating characteristic curve (AUROC) = 0.960, area under the precision-recall curve (AUPRC) = 0.854, accuracy = 0.910) (**Figure 3B-C**).

To avoid potential overfitting, we designed a leave-one-cell line-out cross-validation to test the ProteinTalks model (see model setting 2 in **Methods**). For the majority of cell lines (16 out of 18), the AUROC values were at least 0.9. Additionally, 16 out of 18 cell lines demonstrated the AUPRC values of at least 0.8 (**Figure S4**).

The chemical structure space is almost infinite; therefore, it is crucial to determine whether our model works in chemicals absent in the training data. We then performed leave-one-drug-out cross-validation for the ProteinTalks model (see model setting 3 in **Methods**). We stratified the drugs into four classes based on their accuracy, AUPRC, and AUROC values computed by ProteinTalks (**Figure S5A**). The first class of drugs, which includes 14 drugs, showed relatively high accuracy and high AUPRC/AUROC values. The second class, comprising 21 drugs, exhibited relatively high accuracy but low AUPRC/AUROC values, due to an imbalance in the numbers of ineffective and effective drugs for each cell line. Specifically, the ratio of ineffective to effective among these 21 drugs was significantly higher compared to the other 42 drugs (**Figure S5B**). The third class, including 15 drugs, showed relatively low accuracy but high AUPRC/AUROC, suggesting that while the model performs well, setting the classification threshold of the predicted score at 0.5 may not be optimal. For instance, the accuracy of toremifene citrate efficacy prediction across different cell lines decreased as the threshold of predicted score increased from 0 to 0.25, and then remained at zero for thresholds above 0.25 (**Figure S5C**). The last class, consisting of 13 drugs, displayed the worst performance with low accuracy and low AUPRC/AUROC values, likely due to the absence of drugs with similar MOAs in the training set. For example, three drugs, namely sonidegib diphosphate, irinotecan hydrochloride, and pemetrexed disodium hydrate, modulate protein targets (SMO, TOP1, and DHFR, respectively) that are absent in the training set.

We further generated a dataset to independently assess the model (**Figure S3**, model setting 3 in **Methods**). Here, we extended the perturbation to 98 additional anti-cancer small-molecular compounds, with the baseline proteomic data without perturbation and corresponding drug efficacy data. These compounds were used to treat four TNBC cell lines included in the pretrained dataset. We acquired 5138 drug-dose response measurements (**Figure S3**). For the new compounds in the dataset, the overall accuracy was 0.619, with an AUROC of 0.671. After excluding the drugs with MOA categories absent in the training set, both accuracy and AUROC increased to 0.844 and 0.840, respectively. Remarkably, the accuracies for drug MOA categories present in pretrained datasets were consistent, as illustrated in **Figure S5D**. This consistency across the two test datasets suggests the model's generalization ability in predicting drug efficacy. The data collectively show that ProteinTalks can well predict the drugs with similar MOAs to those in the training set, while for drugs with distinct MOAs absent in the training set, it could still achieve an overall accuracy between 0.6-0.7.

241 **Prediction of drug synergy**

Next, we explored whether ProteinTalks could predict drug synergies. Our pretrained datasets covered 914 combination-cell-line tuples, which have also been investigated by the Garnett group at the Wellcome Sanger Institute¹⁶. The ProteinTalks model, taking the structure information of each drug pair as input, is capable of predicting the synergy of drug combination, as evaluated by the Garnett group's drug synergistic effect (**Figure 3A**). Our model showed that the synergistic scores of synergistic pairs were significantly higher than those of non-synergistic pairs (**Figure 4A**). Among the top 1000 tuples, the majority of potent synergies (58.7%) were observed in combinations of targeted drugs. The next significant proportion (26.2%) consisted of combinations of targeted drugs with chemotherapies.

Combinations of targeted drugs with other types of drugs (12.3%) followed, while the least prevalent synergies (2.8%) were observed between targeted drugs and hormonal agents (**Figure 4B**). These findings suggest that combining targeted drugs with other agents tends to result in synergistic effects. Then we experimentally validated the synergistic efficacy of four synergistic tuples including bosutinib-tucatinib-HCC1143, bosutinib-abemaciclib-HCC70, bosutinib-tucatinib-HCC1395, and bosutinib-abemaciclib-HCC1806 using CCK-8-based cell viability assays. The assays employed a fixed concentration of one drug and a discontinuous 10,000-fold (seven points) dose-response curve of the other drug, following the methodology described previously¹⁶. According to the synergistic screening criteria¹⁶ (**Figure 4C**, $\Delta E_{\max} \geq 0.2$ or $\Delta \log_2(IC_{50}) \geq 3$), four of the top 20 predicted synergistic tuples demonstrated synergy, including the combination of bosutinib and tucatinib in HCC1395 ($\Delta E_{\max} = 0.31$), bosutinib and tucatinib in HCC1143 ($\Delta E_{\max} = 0.28$), bosutinib and abemaciclib in HCC70 ($\Delta E_{\max} = 0.39$), bosutinib and abemaciclib in HCC1806 ($\Delta E_{\max} = 0.74$), and were confirmed to be synergistic (**Figure 4C-D**). These results support the capability of ProteinTalks in drug synergy prediction.

ProteinTalks prioritizes pathways and proteins networks responsible for drug resistance

To investigate whether ProteinTalks could identify the pathways or proteins associated with drug efficacy, we calculated the SHapley Additive exPlanations (SHAP) values to determine the importance of the proteins in the ProteinTalks predictions through comparing effective and ineffective groups based on individual drugs or drug combinations. A higher SHAP value suggests a stronger positive correlation between protein expression level and drug sensitivity. Firstly, we computed the average SHAP values associated with 41 hallmark pathways involving the proteins incorporated in the ProteinTalks models (**Table S4A**). Drugs with similar MOAs exhibited similar SHAP values in these pathways (**Figure 5A**). Pathways with high SHAP values for drug or drug class could be a potential indicator of drug responsiveness. For example, DNA repair pathways are highlighted as vital for predicting the sensitivity to alkylating agents, while the PI3K-AKT-mTOR signaling pathways are central to interpreting responses to PI3K and AKT inhibitors (**Figure 5A**). Similarly, the estrogen response pathway takes precedence in the context of aromatase inhibitors, which block estrogen or androgen production, underscoring its role in the drug MOAs (**Figure 5A**).

We further extended the calculation of SHAP values to drug combinations, thereby enhancing the model's interpretability. Within these combinations, the DNA damage and repair pathways played a critical role in accurately predicting synergy for chemotherapeutic combinations with other agents. Meanwhile, several signaling pathways, including PI3K-AKT-mTOR, TGF- β , and NOTCH, showed a high correlation with the responses to targeted therapies combined with other agents (**Figure S6**).

To evaluate the significance of these prioritized proteins in predicting drug sensitivity, we ranked the average SHAP value for each protein across different drug types. The top 30 proteins with the highest average SHAP values and the bottom 30 proteins with the lowest average SHAP values across all cell lines and drugs are displayed in **Figure S7**, providing a focused snapshot of the most important features. Among these, we observed numerous proteins typically targeted in breast cancer therapies, such as CDK4, CDK6, ERBB2, SRC,

293 mTOR, and TOP2A, thereby validating the biological relevance of our model (**Figure S7**).

294 In addition to these established relevant proteins, our analysis also brought to light new ones.

295 For instance, aldo-keto reductase 1C3 (AKR1C3) emerged as the most important protein for

296 hormonal agents and the second-most important for kinase inhibitors (**Figures S7A-B**).

297 Remarkably, AKR1C3 had the highest SHAP value in MDA-MB-453 cells treated with

298 toremifene, an ER modulator, and also ranked first in terms of SHAP value in the context of

299 treatment with afatinib, an EGFR inhibitor (**Table S4B**). AKR1C3 contributes to cell

300 proliferation and differentiation by significantly enhancing the estrogen biosynthetic pathway

301 ²⁵. Past studies have indicated that AKR1C3 overexpression may reduce TNBC cell sensitivity

302 to doxorubicin ²⁶. Among all samples treated with hormonal agents, the highest SHAP value

303 of AKR1C3 ranking is MDA-MB-453 treated with toremifene (ER modulator) (**Table S4**).

304 We further knocked down the expression of AKR1C3 using siRNA in the MDA-MB-453 cell

305 line from two sources (**Figure 5D**). The efficacy of the siRNA knockdown experiments was

306 confirmed with mass spectrometry analysis (**Figures S7D-E**). Cytotoxicity assays revealed

307 that AKR1C3 knockdown enhanced the sensitivity of MDA-MB-453-1 cells to toremifene

308 ($\Delta\log_2(\text{IC}_{50}) = 4.76$) and MDA-MB-453-2 cells to afatinib ($\Delta\log_2(\text{IC}_{50}) = 3.32$) (**Figure**

309 **5C**).

310 Within the alkylating agents category, TYMS was identified as the third most significant

311 protein (**Figure S7C**), with its role in drug sensitivity corroborated in **Figures S2B-C**.

312 Another protein involved in nucleic acid biosynthesis, CMPK1, ranked seventh. Knockdown

313 of CMPK1 in HCC70 cells (**Figure S7F**) led to enhanced sensitivity to the kinase inhibitors

314 decitabine ($\Delta\text{Emax} = 0.24$) and azacytidine ($\Delta\text{Emax} = 0.26$) (**Figure 5C**). In summary, the

315 ProteinTalks model effectively uncovers interpretable proteins related to drug resistance.

316 **Finetuning allows drug response prediction based on transcriptome profiles of**

317 **patient-derived xenografts in mice**

318 Next, we explored whether the drug sensitivity prediction could be extended to breast cancer

319 patient-derived tumor cells (PDTCs). We referred to a recent study of 30 short-term cultured

320 PDTCs from patient-derived tumor xenograft (PDTX) models, treated with 96 compounds ²⁷.

321 Baseline transcriptome profiles of these PDTCs were acquired in this study. In the first-stage

322 training, the ProteinTalks model was trained on varying ratios of the perturbation proteomic

323 data (PTDS1-3 datasets), ranging from 0% to 90%. To address the differences between PTDS

324 and PDTC datasets, we then implemented finetuning to build a multi-omics model, model-1.

325 This process involved utilizing the 25 randomly selected baseline transcriptome profiles of

326 PDTCs, phenotypic data of the 25 PDTCs' response to 96 drugs, as well as drug information

327 of 881 DMF, 55 DPP, and 40 targets (**Figure S8A**). As the PDTC dataset only provided drug

328 efficacy data and lacked perturbed transcriptomic data, training Loss₁ for model-1 was not

329 feasible. The finetuning was conducted using Loss₂ only (**Figure S8A**). For comparison, we

330 trained the model-1 from scratch solely on the same subset of transcriptomic PDTC data

331 without finetuning from the perturbation proteomic data, named model-1-wo (median

332 accuracy: 0.786, median AUROC: 0.888, median AUPRC: 0.926). When 90% of the

333 pretrained datasets were used in the first stage, the corresponding models obtained were

334 model-1-90% (**Figure S8A**). The model-1-90% exhibited a significant enhancement in

335 predicting of drug efficacy (median accuracy: 0.819, median AUROC: 0.918, median AUPRC:

336 0.951) (**Figure 6A, S8B**).

337 We also implemented similar finetuning on another 1075 pan-cancer PDX models'
338 transcriptomic dataset to predict drug response²⁸. The median accuracy improved from 0.613
339 to 0.736 when comparing model-1-wo to model-1-90% (**Figure 6B**). This suggests the
340 generalizability and adaptability of ProteinTalks in enhancing transcriptomic data for
341 predicting therapeutic outcomes based on protein network dynamics.

342 **Application of ProteinTalks to clinical biopsy specimens**

343 To further validate this approach using clinical biopsy tissue samples, we tested the drug
344 response dataset in different terms. We firstly evaluated short-term drug response using
345 baseline transcriptomic data from 16 clinical TNBC patients prior to receiving neoadjuvant
346 carboplatin and docetaxel combination chemotherapy²⁹. Since our model can select core
347 proteins associated with cellular response to specific categories of drugs, we assessed the
348 effectiveness of the top 60 proteins screened by ProteinTalks in predicting chemotherapy
349 sensitivity in these biopsy tissue samples. Specifically, the prognostic performance of these
350 top 60 proteins was compared to that of the entire proteome and a randomly selected subset of
351 60 proteins. The results showed that the selected proteins significantly outperformed the
352 random subsets in predicting chemotherapy drug sensitivity in TNBC and showed comparable
353 or superior performance to the full protein set across multiple methods, including bootstrap,
354 random forest, logistic regression, SGD, and KNN (**Figures S8C-E**).

355 We then performed long-term survival prognosis prediction in two public datasets: a
356 transcriptomic dataset of 823 BC patients from TCGA (<https://xenabrowser.net/>) and a
357 proteomic dataset of 122 BC patients from CPTAC³⁰. For these patients, who received
358 various chemotherapy or targeted treatments in these datasets, we selected the top 60 proteins
359 identified by ProteinTalks (**Table S5**), which represent the most affected by all categories of
360 drugs. These proteins distinguished the patients into two groups, a high-risk group with
361 poorer prognosis and a low-risk with better prognosis, in both TCGA (**Figure S9**) and
362 CPTAC (**Figure 6C-D**) datasets.

363 These findings suggest that the ProteinTalks model effectively identifies clinically relevant
364 biomarkers. It could improve the prediction of drug responses using baseline biopsies from
365 TNBC patients prior to treatment, thereby potentially streamlining the pursuit of precision
366 therapy.

367 **Discussion**

368 Understanding the mechanisms and interactions within cells is a fundamental question in
369 biology. To advance this, we generated a large-scale proteomic perturbation dataset and
370 developed a corresponding dynamical foundation model named ProteinTalks, which also
371 serves as a milestone towards constructing a virtual cell. The over 38 million protein
372 measurements from 16,000 perturbed proteomic samples that we produced in-house makes it
373 the largest proteomics dataset to date. In comparison, previous large-scale perturbation
374 proteomics studies^{8,10,11,31} were at least ten times smaller in size, insufficient to be used
375 directly for building AI models.

376 Based on the large-scale temporal perturbation data, we developed a dynamical foundation

model framework, which we believe is the first in its kind. core of the ProteinTalks model lies in describing the dynamic changes of variables over time, capturing the time-dependent behavior of the cell systems. This capability enables the model to gain a comprehensive understanding of protein network dynamics and can be applied in multiple downstream tasks. This model achieves an AUROC that is 5% to 27% higher than other models in predicting the efficacy of both single and combination therapies. Furthermore, we conducted validation experiments to confirm the synergy of four drug combinations. To delve deeper, we employed gene interference technology to confirm the involvement of specific proteins in drug response. For example, AKR1C3 was found to play a role in the response to toremifene and afatinib, while CMPK1 was found to contribute to the effects of decitabine and azacytidine. It creates a distinct latent space for clear differentiation between cellular states and drug types. Additionally, ProteinTalks can identify critical proteins and pathways associated with drug efficacy. For instance, DNA repair pathways play a significant role in predicting sensitivity to alkylating agents and chemotherapeutic combinations with other drugs. Similarly, the PI3K-AKT-mTOR signaling pathways are essential for interpreting responses to PI3K and AKT inhibitors, as well as other targeted therapies when combined with other drugs. Through finetuning, this model enables predictions of the drug efficacy for patient-derived tumor xenograft (PDX) models. Furthermore, ProteinTalks can identify biomarkers associated with responses in 16 TNBC patients receiving neoadjuvant chemotherapy. Applied to multi-omics data of tumor samples collected before therapeutic treatment from TCGA and CPTAC, ProteinTalks can identify prognostic biomarkers in breast cancer patients, suggesting the potential of ProteinTalks to forecast treatment outcomes of diseases. Therefore, ProteinTalks represents a foundation model which has learned sophisticated protein network dynamics, and can be utilized to a broad range of downstream tasks.

Numerous AI models have been developed to map complex molecular networks for virtual cell construction. Among these, large-scale protein structure data have been used to build AlphaFold^{32,33} for precise protein structure and interaction prediction. Single-cell RNA-seq data corpus have led to the development of multiple foundation models for cellular status prediction, such as Geneformer³⁴ and scGPT³⁵. However, there is no proteome foundation model for predicting protein network dynamics due to the limited availability of proteomic data, which constrains the application of AI in this context. Therefore, this is the first proteome foundation model, with dynamical information. Nevertheless, this study has several limitations. The proteome depth for each sample is less than several other recent papers^{8,10,11}. This is because we intentionally chose a high-throughput DIA-MS methodology of relatively low cost for this study (~70 RMB or 10 \$ per proteome), otherwise, it won't be feasible for a laboratory to acquire 16,000 perturbed proteomes to test whether this number of perturbations is sufficient for building a foundation model. Post-translational modifications (PTMs) are crucial for drug actions; however, we were not able to include them in this pilot study. Nevertheless, we have set up the framework of dynamical proteomic study in the context of foundation model and demonstrated its power in various tasks, paved way for future studies to put more efforts to identify more proteins, PTMs and drug combination perturbed proteomic data.

419

420 **Materials and Methods**

421 **Cell line panel**

422 BT20, BT549, DU4475, HCC1143, HCC1395, HCC1806, HCC1937, HCC38, HCC70,
423 Hs578T, MDA-MB-436, MDA-MB-453-1, MDA-MB-468, HCC1187, T47D, and MCF7 cell
424 lines were purchased from ATCC, while MDA-MB-453-2 and MDA-MB-231 were purchased
425 from Meisen. The detailed information is listed in **Table S1A**. We have MDA-MB-453 cell
426 lines from two sources. To distinguish them, the one from ATCC is denoted as
427 MDA-MB-453-1, while the one from Meisen as MDA-MB-453-2.

428 **Cell viability assay**

429 To determine the half maximal inhibitory concentration (IC₅₀) in primary screening, we
430 employed the CCK8 (Sigma) cell proliferation assay in accordance with the manufacturer's
431 protocol. Log-phase cells (**Table S1A**) were inoculated into a 96-well plate at a volume of
432 100 µL per well, using basal medium supplemented with 10% fetal bovine serum (FBS). After
433 a 24-hour incubation, the medium was replaced with a gradient of drug concentrations (0.5, 5,
434 50, 200 µM) in medium containing 5% FBS, followed by another 24-hour incubation period
435 (**Table S1C**). Cells without drug treatment served as negative controls. DMSO levels were
436 ensured not to exceed 0.5%. Wells lacking cells acted as blank controls. Subsequently, 10 µL
437 of CCK8 reagent (5 mg/mL) was added to each well, and the plates were incubated for 4
438 hours at 37°C. Absorbance was measured at 450 nm using a Multiscan Spectrum (BioTek,
439 USA) to calculate the IC₅₀ values. The criteria for drug efficacy were as follows:

440 (1) Cell viability after drug treatment should increase with decreased drug concentrations
441 (variance <20%) to be considered valid data. If cell viability rates treated with 0.5, 5, and 50
442 µM drug concentrations all exceed 50%, the drug is deemed ineffective.

443 (2) If cell viability at 50 µM is not only less than 50% but also shows a further reduction
444 to below 50% of cell viability observed at 0.5 µM, the drug is deemed effective, and vice
445 versa.

446 Based on these criteria, we identified effective drugs and subsequently tested them across nine
447 gradient concentrations over a 72-hour duration.

448 **Drug treatment for proteomics preparation**

449 For cell culture and drug treatment, each 96-well plate included a control sample (no drug
450 treatment) and a blank sample (no cells, medium only) for comparison. Samples within a
451 single 96-well plate were treated as a batch. Triplicate biological replicates were performed
452 for each treatment condition. The efficacy of each drug (IC₅₀) was assessed bi-monthly on a
453 minimum of two cell lines to ensure reproducibility. In single-drug experiments, a standard
454 concentration of 10 µM was selected for perturbation, drawing on precedents from large-scale
455 drug screening studies^{8,36,37}. A comprehensive assay of 2025 drug combinations¹⁶ informed
456 the concentrations used in combination treatments. Following drug treatment, cells were
457 washed three times with PBS and lysed in lysis buffer containing 6 M urea, 2 M thiourea, and
458 10 mM ammonium bicarbonate buffer (ABB). Proteins were then reduced and alkylated using
459 TCEP and IAA, respectively, followed by digestion with trypsin.

460 **LC-MS/MS analysis**

461 For the proteomics analysis, we injected 1 µg of purified peptides into an Eksigent Nano LC
462 415 system (with a 1-10 µL/min flow module to switch the LC from nano-flow to
463 micro-flow). Chromatographic separation was achieved using an Eksigent analytical column
464 (C18 ChromXP, 3 µm, 0.3 x 150 mm) and trap column (C18 ChromXP, 5 µm, 0.3 x 10 mm),
465 as detailed in prior documentation³⁸. Mobile phase Buffer A consisted of 2% acetonitrile
466 (ACN) with 0.1% formic acid (FA), and Buffer B comprised 98% ACN with 0.1% FA. A
467 linear gradient of 5-32% Buffer B was applied over 15 minutes at a flow rate of 5 µL/min.
468 The coupled DIA-MS analysis was conducted on a TripleTOF 5600+ system, with the ion
469 accumulation time set at 150 ms for MS1 (m/z 350-1250) and 20 ms for each DIA window.
470 We optimized the DIA window scheme to include 71 variable windows and operated the
471 instrument in high sensitivity mode.

472 To generate the spectral library, we fractionated 10 mg of peptides derived from various
473 TNBC cell lines using the Ultimate 3000 system, following established protocols³⁹. The
474 resultant TNBC-specific spectral library encompassed 130,130 peptides corresponding to
475 9355 proteins.

476 In DIA-MS analysis, a pooled sample of nine TNBC cell lines (BT20, BT549, HCC1143,
477 HCC1395, HCC1806, HCC1937, HCC38, HCC70, Hs578T), along with randomly chosen
478 technical replicates, were employed as quality control samples within each batch.

479 Raw DIA data files were initially converted to profile mode mzML format using msConvert.
480 Subsequent analysis of mzML files was performed with DIA-NN software (version 1.7.15),
481 as previously described^{40,41}. Analysis parameters were meticulously set: trypsin/P as the
482 protease with a maximum of two missed cleavages allowed; N-terminal methionine excision,
483 cysteine carbamidomethylation, and methionine oxidation as the specified modifications;
484 peptide length restricted to a range of 7 to 30 amino acids; and an m/z range of 400 to 2000
485 for precursor ions and 100 to 2000 for fragment ions. False discovery rate (FDR) cutoffs for
486 precursor ions, peptides, and proteins were stringently maintained at 1%. Data were processed
487 using the single-pass mode of the neural network classifier.

488 **Proteomics data preprocessing and normalization**

489 From the initial dataset, 689 samples with identification counts below 1000 were discarded,
490 leaving 16,311 samples for analysis. No proteins exhibited a missing rate greater than 90% in
491 the complete dataset, with an overall missing rate of 51.7%. The initial step in data
492 preprocessing involved imputing missing values with a value equivalent to 0.8 times the
493 minimum detected intensity. The reproducibility of biological replicates, technical replicates,
494 and pooled samples was assessed using Pearson correlation and the coefficient of variation
495 (CV) as metrics.

496 **Differentially expressed analysis**

497 Prior to differential expression analysis, proteins that were absent in more than 80% of the
498 samples across both groups were excluded. The comparison between the two groups was
499 conducted using a two-sided unpaired Welch's t-test. P-values obtained from the statistical
500 tests were adjusted for multiple comparisons using the Benjamini-Hochberg (B-H) method.

Differentially expressed proteins were identified based on a combination of fold change and either raw p-values or B-H adjusted p-values.

PertScore calculation

The PertScore quantifies the cumulative frequency of protein expression alterations induced by various drugs across different time points in each cell line, aiming to highlight the most recurrent protein expression changes triggered by drug perturbations. Proteins were scored based on their perturbation impact and visualized in a three-dimensional coordinate system (Figure S2A), where the x-axis represents different cell lines, the y-axis denotes drug types, and the z-axis indicates drug treatment durations. Each coordinate (Xl, Ym, Zn) signifies the outcome of the differential analysis between samples from cell line l treated with drug type m at time point n and their corresponding untreated controls. For a given coordinate, if a protein is significantly upregulated, it is assigned a score of 1. Conversely, a significant downregulation is scored as -1. Proteins that do not exhibit significant changes are assigned a score of 0.

Pathway enrichment

To elucidate the biological pathways associated with the differentially expressed proteins, we employed four well-established databases: KEGG pathway⁴², Metascape⁴³, STRING⁴⁴, and Ingenuity Pathway Analysis⁴⁵ (IPA, version 51963813). Pathway enrichment analysis was performed with a defined significance threshold of $p < 0.01$, and only pathways containing at least two proteins or metabolites from our dataset were considered for further analysis.

mFuzz analysis

One-way analysis of variance (ANOVA) was used to determine differences between samples treated at different time points ($p < 0.05$). The average normalized protein quantities, z-score in each GS grade were used for fuzzy c-means clustering with the R (version 4.0.2) package Mfuzz (version 2.48.0). The number of clusters was set to four, and the fuzzifier coefficient, M, was set to 1.25. The clusters with consistently increasing or decreasing trends were selected. In **Figure 2B**, the clustering was based on the drug group, while in **Figure 2C**, the clustering was based on both the drug group and drug sensitivity. The proteins that overlapped between the increasing clusters in the sensitive group and the decreasing clusters in the resistant group were filtered out. This filtration process is depicted in the upper panel of **Figure 2C**, and the same procedure was conducted in reverse for the other group.

Data preprocessing

After obtaining data from different cell lines, various drug perturbations, and multiple time points, we averaged the samples from the repeated experiments, for example, data for 6 hrs in HCC70 with drug #75. Prior to inputting the data into the model, we performed min-max normalization on each sample. Ultimately, this process yielded the expression levels of 5585 proteins under each perturbation condition, across different cell lines, at 0, 6, 24, and 48 hours.

SMILES generation

Using the SMILES representations of the drug molecules, we employed the R-package ChemmineR⁴⁶. We set the parameters as follows, functions MW and MF parameter

addH=FALSE, function bonds parameter type="charge", function groups parameter type="countMA", function rings parameters upper=6, type="count", arom=TRUE. We obtained the 881-dimensional drug molecular fingerprints (DMF) and the 55-dimensional drug physicochemical properties for each drug.

Data partitioning for model performance evaluation

We evaluated the model's performance under three distinct datasets with different partitioning settings:

Setting 1: Model Performance Test

All time points are treated as a combined sample. The dataset, including various cell lines and drug perturbations, is randomly divided into training, validation, and test sets with a split ratio of 0.7:0.2:0.1, resulting in 1070, 305, and 154 samples respectively.

Setting 2: Model Performance Test for Cross-Cell-Type Data

One cell line is left out at a time. The model is trained and validated on data from the remaining cell lines and tested on the excluded cell line, assessing cross-cell-type performance.

Setting 3: Model Performance Test with Individual Drugs

Tests the model on previously unseen drugs using the results from Setting 1. This assesses the model's generalizability to new drug perturbations.

These settings ensure a comprehensive evaluation of the model's robustness and generalizability in predicting protein responses to drug perturbations.

Model architecture and parameters

The initial component of our model is devoted to predicting post-perturbation protein expression using pre-perturbation protein expression as input data. Initially, the model ingests data from 5,585 proteins P_0 , concatenating this with corresponding dimension perturbation data D . This is processed through a linear layer L_1 , which elevates the dimensionality from 2 dimensions to 32 dimensions:

$$L_1(P_0, D) = W_1 \cdot [P_0, D] + b_1, \#(1.)$$

where W_1 is the weight matrix and b_1 is the bias vector.

Subsequently, a convolutional network C_1 is employed to further enhance the dimensionality to 128 dimensions. Following this, we have engineered a two-layer linear network, utilizing the Softplus as the activation function and implementing a dropout ratio of 0.1. The output of the convolutional network is passed through two linear layers L_2 and L_3 :

$$L_2(C_1) = \text{Softplus}(W_2 \cdot C_1 + b_2), \#(2.)$$

$$L_3(\text{Dropout}(L_2)) = \text{Softplus}(W_3 \cdot \text{Dropout}(L_2) + b_3), \#(3.)$$

The next step involves parameterizing this two-layer linear network with Neural Ordinary Differential Equations (Neural ODEs), opting for the 'rk4' solver, a fourth-order Runge-Kutta method, to predict omics data over multiple time points:

$$\text{Neural_ODE}(L_3) = \text{ODESolve}(L_3, \text{rk4}), \#(4.)$$

Then, a convolutional network C_2 is applied to reduce the dimensionality of the predicted multi-time point embeddings, decoding them down to 32 dimensions. Finally, a linear layer

579 L_4 is utilized to decode the 32-dimensional representation back into the proteomics space,
580 and get predicted proteomics $\tilde{P}_{6hrs}, \tilde{P}_{24hrs}, \tilde{P}_{48hrs}$:

$$\tilde{P}_{6hrs}, \tilde{P}_{24hrs}, \tilde{P}_{48hrs} = L_4(C_2) = W_4 \cdot C_2(\text{Neural_ODE}(L_3)) + b_4. \#(5.)$$

581 By utilizing an MSE loss, the first part of ProteinTalks denoted a loss function as Loss_1 :

$$\text{Loss}_1 = \text{MSE}([\tilde{P}_{6hrs}, \tilde{P}_{24hrs}, \tilde{P}_{48hrs}], [P_{6hrs}, P_{24hrs}, P_{48hrs}]), \#(6.)$$

582 where $P_{6hrs}, P_{24hrs}, P_{48hrs}$ is the ground-truth of proteomics at time 6h, 24h, 48, respectively.

583 The second part of our model is centered around the prediction of drug efficacy. This involves
584 concatenating the proteomics data $\tilde{P}_{6hrs}, \tilde{P}_{24hrs}, \tilde{P}_{48hrs}$ predicted at specific or all time
585 points with the initial proteomics data P_0 , followed by a convolutional network C_3 that
586 elevates the data dimensionality to 128 dimensions:

$$C_3([P_0, \tilde{P}_{6hrs}, \tilde{P}_{24hrs}, \tilde{P}_{48hrs}]) \rightarrow 128 \text{ dimensions.} \#(7.)$$

587 To describe the physicochemical properties of small molecule drugs, we utilize descriptions
588 based on Simplified Molecular-Input Line-Entry System (SMILES), obtaining 935 features.
589 These drug features for two drugs D_1 and D_2 (or duplicate if only one drug was used) are
590 concatenated, forming a tensor $[D_1, D_2]$ with a dimensionality of 935×2 , which is then
591 expanded to 128 dimensions through another convolutional network C_4 :

$$C_4([D_1, D_2]) \rightarrow 128 \text{ dimensions.} \#(8.)$$

592 Subsequently, the proteomics data and drug features are concatenated. This combined dataset
593 is processed through a linear layer L_5 utilizing the ReLU activation function to reduce its
594 dimensionality to 32 dimensions:

$$L_5([C_3, C_4]) = \text{ReLU}(W_5 \cdot [C_3, C_4] + b_5). \#(9.)$$

595 Finally, the model was followed by a linear layer, applying a sigmoid function to the output,
596 which yields a prediction on the efficacy of the drug or drug combo synergy $\tilde{S}$:

$$\tilde{S} = L_6(L_5) = \text{sigmoid}(W_6 \cdot L_5 + b_6). \#(10.)$$

597 By utilizing a binary cross entropy (BCE) loss, the second part of ProteinTalks denoted a loss
598 function as Loss_2 :

$$\text{Loss}_2 = \text{BCE}(\tilde{S}, S), \#(11.)$$

599 where S is the ground-truth of drug efficacy or combo synergy.

600 Final loss function for ProteinTalks is:

$$\text{Loss} = (1 - \lambda)\text{Loss}_1 + \lambda\text{Loss}_2, \#(12.)$$

601 where λ is the hyperparameter for balancing the importance of the two tasks. The λ was set
602 to 0.8 for ProteinTalks.

603 Multi-task learning

604 Given that ProteinTalks is a multi-task learning task, we address the gradient conflicts
605 between the losses from different tasks as Loss function (12), specifically Loss1 and Loss2,
606 by calculating their cosine similarity. The cosine similarity of the gradients is computed as
607 follows:

$$\text{cosine_similarity} = \frac{\nabla \text{Loss}_1 \cdot \nabla \text{Loss}_2}{\|\nabla \text{Loss}_1\| \|\nabla \text{Loss}_2\|} \quad \#(13)$$

608 We then apply a clipping operation using a cutoff value of 1.0:

$$\text{clipped_similarity} = \min(\text{cosine_similarity}, 1.0) \quad \#(14)$$

609 Based on the clipped cosine similarity, we adjust the weights of the losses by a factor of 0.01:

$$\text{adjustment_factor} = 0.01 \times \text{clipped_similarity} \quad \#(15)$$

610 The total loss is updated by modifying the weight of each task based on the gradient
611 similarities. If the gradient directions are similar or aligned, we retain the original weights. If
612 the gradient directions conflict, we prioritize the drug efficacy prediction task by increasing
613 its gradient weight while decreasing the weights of other tasks. The adjustment is done as
614 follows:

615 If gradients are similar:

$$w_{\text{Loss}1} = w_{\text{Loss}1}, w_{\text{Loss}2} = w_{\text{Loss}2} \quad \#(16)$$

616 If gradients conflict:

$$w_{\text{Loss}1} = w_{\text{Loss}1} - \text{adjustment_factor} \quad \#(17)$$

$$w_{\text{Loss}2} = w_{\text{Loss}2} + \text{adjustment_factor} \quad \#(18)$$

617 Finally, we ensure that the weights remain within a reasonable range. The updated gradient is
618 then backpropagated through the model to update the parameters. For implementation details,
619 refer to the code.

620 **Implementation of relevant methods for benchmarking**

621 By concatenating data from multiple time points and combining it with drug data, we
622 modified the input data dimensions of the DeepSynergy code to fit our dataset. DeepSynergy
623 is a fully connected network that integrates omics data and drug information into the model to
624 predict the efficacy of drug combinations. Other parameters were kept as original. We then
625 applied DeepSynergy to our data. For Bootstrap, we used Bagging with a Decision Tree,
626 implemented with *scikit-learn*'s *BaggingClassifier* function, setting parameters as
627 *n_estimators=100* and *bootstrap=True*. For Random Forest, we used *scikit-learn*'s
628 *RandomForestClassifier* function, with default parameters from version 1.3.0 of *scikit-learn*.
629 For Logistic Regression, we used *scikit-learn*'s *LogisticRegression* function, setting the
630 parameter *max_iter=1000*. For stochastic gradient descent, we used *scikit-learn*'s
631 *SGDClassifier* function, with the parameter *max_iter=1000*. For the K-nearest neighbor
632 algorithm, we used *scikit-learn*'s *KNeighborsClassifier* function, with default parameters from
633 version 1.3.0 of *scikit-learn*.

634 **SHAP value calculation**

635 The significance of individual proteins in relation to sensitivity of single drug or the synergy
636 of drug combination was determined using SHAP (SHapley Additive exPlanations) values for
637 Loss_2 ⁴⁷. The analysis commenced by calculating the SHAP value for each protein, thereby
638 assessing its specific contribution to the predictive model's output. For each cell line and drug

condition, the SHAP values of each protein were computed for sensitive drug or effective drug combinations, using the resistant drugs or no synergy combinations from other cell lines or drugs as background samples. This evaluation framework permitted the assessment of particular proteins' roles in predicting the effectiveness of drugs or their combinations.

SHAP value filtration

Proteins were then ranked according to their average SHAP values across all samples. The top 30 proteins with the most significant positive SHAP values were identified as having a favorable influence on sensitivity of single drug or the synergy of the drug combination, while the 30 proteins with the highest negative SHAP values, in absolute terms, were recognized as having an adverse effect. This selection of 60 proteins represents the most influential factors, thus forming the core set of features with the most substantial impact on the efficacy of single drugs or drug combinations.

By leveraging SHAP values in this manner, we pinpointed key proteins that modulate the response to single drugs or drug combinations, enhancing our understanding of their mechanistic roles and informing the refinement of predictive models for drug sensitivity or drug combination synergy.

Validation of clinical data for top 60 proteins by SHAP values

After model training, we selected the top 60 proteins with the highest SHAP values for antimitotic drugs. For 16 individual clinical samples, we conducted 4-fold cross-validation and repeated the process 100 times for each method. For Bootstrap, we used *scikit-learn's* *BaggingClassifier* function with the parameter *n_estimators=10*. For Random Forest, we used *scikit-learn's* *RandomForestClassifier* function with the parameters *n_estimators=100* and *criterion='log_loss'*. For Logistic Regression, we used *scikit-learn's* *LogisticRegression* function with the parameter *max_iter=1000*. For stochastic gradient descent, we used *scikit-learn's* *SGDClassifier* function with the parameters *max_iter=1000*, *tol=1e-3*, and *loss='log_loss'*. For the K-nearest neighbor algorithm, we used *scikit-learn's* *KNeighborsClassifier* function with the parameter *n_neighbors=3*.

Transfer learning for the transcriptomics data of PDTC model

To illustrate the complementary nature of proteomics data to transcriptomics data, we conducted a study comparing the effects of models pretrained on proteomics data versus those trained from scratch on transcriptomics data^{27,28}. We employed various proportions of proteomics data to train the ProteinTalks models in the first stage, specifically using 90% for training and 10% for validation. Subsequently, models trained on proteomics data were transferred to transcriptomics data, utilizing a dataset comprising 90% samples for training and 10% for testing. Since the ground truth only includes drug efficacy data and lacks omics data at future time points, the hyperparameter λ in the loss function was set to 1. Consequently, the loss function for ProteinTalks is defined as:

$$\text{Loss} = \text{Loss}_2.\#(19)$$

After training for 2000 epochs, the ProteinTalks model was obtained. The performance of these models was then assessed using the test dataset to validate the efficacy of leveraging proteomics data to enhance the predictive capabilities of transcriptomics-based models.

679 **Machine learning for the transcriptomics data**

680 We extracted the expression matrix of these 62 proteins from the TCGA-BRCA dataset and
681 CPTAC-BRCA dataset, and applied the train_test_split method of the scikit-learn package in
682 Python to divide the dataset into the training dataset and the test dataset. Random forest,
683 Support vector machine, and Gradient boosting decision tree models are used for training and
684 prediction. The performances of three machine learning models were compared, and random
685 forest showed the best performance on the test dataset.

686 For each model, we first built a GridSearchCV parameter search and then trained a
687 classification model to predict alive/death. All models were trained using 3-folds cross
688 validation in the train dataset (N =658) and tested against the test dataset (N =165). We found
689 that random forest showed significantly better performance in independent validation sets.
690 For the random forest model, our parameter adjustment range is: max_depth: [None, 2, 3],
691 min_samples_leaf: [2, 4], min_samples_split: [2, 4, 8], n_estimators: [20, 30, 50, 100, 500,
692 1000]. The accuracy and AUPRC of the model on the test set are output.

693 **Survival analysis for the BC patients**

694 Survival outcomes were defined using time-to-event data, with death as the endpoint and a
695 binary incidentor for the event occurence. For the Cox model in the survival package, all 62
696 proteins were used as variables, and Cox multivariate analysis was performed to analyze the
697 relationship between multiple proteins and sample survival in R. The Cox model calculated
698 the coefficient value of each protein, which was then multiplied by the corresponding protein
699 expression in each sample, and the average value was calculated as the risk score. The risk
700 scores were divided into grouping variables using the surv_cutpoint function of survminer
701 package. The Survfit function was then used to generate Kaplan-Meier survival curves.

702 **Gene interference in different cell lines**

703 Breast cancer cell lines were subjected to gene interference using small interfering RNA
704 (siRNA) in conjunction with Hieff Trans liposomal transfection reagent (Yeasten), following
705 the manufacturer's protocol. The siRNA sequences targeted against AKR1C3 and CMPK1
706 were as follows: for AKR1C3, 5'-GGAACUUUCACCAACAGAU-3'; and for CMPK1,
707 5'-GAGUAGUGGUAGGAGUGAU-3'. Initially, siRNA and transfection reagent were
708 individually mixed with Opti-MEM medium (ATCC) and allowed to incubate for 5 minutes at
709 room temperature. These mixtures were then combined and incubated for an additional 15
710 minutes at room temperature to allow complex formation. The siRNA-liposome complexes
711 were added to the cell suspensions, which were subsequently transferred to 96-well plates for
712 incubation over a 24-hour incubation period. Following transfection, the cells were processed
713 for subsequent experimental steps, which included drug treatment for sensitivity assays and
714 protein harvesting for proteomic analyses, as previously described.

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Author contributions

T.G., H.W. and Y.Z., designed and supervised the project. R.S., L.Q., X.Z., W.H., Q.X., Z.L., and G.Z., conducted proteomic analysis. H.C, Z.X, L.T., and R.S. conducted bioinformatics analysis. Y.L., P.Z., and H.W. performed machine learning analysis. R.S., L.Q., Y.Z., and T.G. interpreted the data with inputs from all co-authors and wrote the manuscript with inputs from co-authors.

Declaration of interests

T.G. and Y.Z. are shareholders of Westlake Omics Inc. L.T., Y.Z., and W.H. are employees of Westlake Omics Inc. H.W. and Y.L. are employees of DP Technology Co., Ltd. The remaining authors declare no competing interests.

References

- 1 Gut, G., Herrmann, M. D. & Pelkmans, L. Multiplexed protein maps link subcellular organization to cellular states. *Science* **361** (2018).
- 2 Cho, N. H. *et al.* OpenCell: Endogenous tagging for the cartography of human cellular organization. *Science* **375**, eabi6983 (2022).
- 3 Johnson, G. T. *et al.* Building the next generation of virtual cells to understand cellular biology. *Biophys J* **122**, 3560-3569 (2023).
- 4 Aebersold, R. & Mann, M. Mass-spectrometric exploration of proteome structure and function. *Nature* **537**, 347-355 (2016).
- 5 Greenblatt, J. F., Alberts, B. M. & Krogan, N. J. Discovery and significance of protein-protein interactions in health and disease. *Cell* **187**, 6501-6517 (2024).
- 6 Kuenzi, B. M. & Ideker, T. A census of pathway maps in cancer systems biology. *Nat Rev Cancer* **20**, 233-246 (2020).
- 7 Meissner, F., Geddes-McAlister, J., Mann, M. & Bantscheff, M. The emerging role of mass spectrometry-based proteomics in drug discovery. *Nat Rev Drug Discov* **21**, 637-654 (2022).
- 8 Mitchell, D. C. *et al.* A proteome-wide atlas of drug mechanism of action. *Nat Biotechnol* **41**, 845-857 (2023).
- 9 Qian, L. *et al.* AI-empowered perturbation proteomics for complex biological systems. *Cell Genom* **4**, 100691 (2024).
- 10 Zecha, J. *et al.* Decrypting drug actions and protein modifications by dose- and time-resolved proteomics. *Science* **380**, 93-101 (2023).
- 11 Eckert, S. *et al.* Decrypting the molecular basis of cellular drug phenotypes by dose-resolved expression proteomics. *Nat Biotechnol* (2024).
- 12 Molinelli, E. J. *et al.* Perturbation biology: inferring signaling networks in cellular systems. *PLoS Comput Biol* **9**, e1003290 (2013).

763 13 Xiao, Q. *et al.* High-throughput proteomics and AI for cancer biomarker discovery. *Adv Drug*
764 *Deliv Rev* **176**, 113844 (2021).

765 14 Gillet, L. C. *et al.* Targeted data extraction of the MS/MS spectra generated by
766 data-independent acquisition: a new concept for consistent and accurate proteome analysis.
767 *Mol Cell Proteomics* **11**, O111 016717 (2012).

768 15 Nyman, E. *et al.* Perturbation biology links temporal protein changes to drug responses in a
769 melanoma cell line. *PLoS Comput Biol* **16**, e1007909 (2020).

770 16 Jaaks, P. *et al.* Effective drug combinations in breast, colon and pancreatic cancer cells. *Nature*
771 **603**, 166-173 (2022).

772 17 Marcotte, R. *et al.* Functional Genomic Landscape of Human Breast Cancer Drivers,
773 Vulnerabilities, and Resistance. *Cell* **164**, 293-309 (2016).

774 18 Li, X. *et al.* LncRNA NEAT1 promotes autophagy via regulating miR-204/ATG3 and
775 enhanced cell resistance to sorafenib in hepatocellular carcinoma. *J Cell Physiol* **235**,
776 3402-3413 (2020).

777 19 Hu, J. *et al.* BTF3 sustains cancer stem-like phenotype of prostate cancer via stabilization of
778 BMI1. *J Exp Clin Cancer Res* **38**, 227 (2019).

779 20 Phi, L. T. H. *et al.* Cancer Stem Cells (CSCs) in Drug Resistance and their Therapeutic
780 Implications in Cancer Treatment. *Stem Cells Int* **2018**, 5416923 (2018).

781 21 De Greve, J. & Giron, P. Targeting the tyrosine kinase inhibitor-resistant mutant EGFR
782 pathway in lung cancer without targeting EGFR? *Transl Lung Cancer Res* **9**, 1-3 (2020).

783 22 Liu, L. *et al.* The LIS1/NDE1 Complex Is Essential for FGF Signaling by Regulating FGF
784 Receptor Intracellular Trafficking. *Cell Rep* **22**, 3277-3291 (2018).

785 23 Park, G. B., Jeong, J. Y., Choi, S., Yoon, Y. S. & Kim, D. Glucose deprivation enhances
786 resistance to paclitaxel via ELAVL2/4-mediated modification of glycolysis in ovarian cancer
787 cells. *Anticancer Drugs* **33**, e370-e380 (2022).

788 24 Ruzzene, M., Bertacchini, J., Toker, A. & Marmioli, S. Cross-talk between the CK2 and AKT
789 signaling pathways in cancer. *Adv Biol Regul* **64**, 1-8 (2017).

790 25 Lin, H. K. *et al.* Expression and characterization of recombinant type 2 3
791 alpha-hydroxysteroid dehydrogenase (HSD) from human prostate: demonstration of
792 bifunctional 3 alpha/17 beta-HSD activity and cellular distribution. *Mol Endocrinol* **11**,
793 1971-1984 (1997).

794 26 Boichuk, S. *et al.* Establishment and characterization of a triple negative basal-like breast
795 cancer cell line with multi-drug resistance. *Oncol Lett* **14**, 5039-5045 (2017).

796 27 Bruna, A. *et al.* A Biobank of Breast Cancer Explants with Preserved Intra-tumor
797 Heterogeneity to Screen Anticancer Compounds. *Cell* **167**, 260-274 e222 (2016).

798 28 Gao, H. *et al.* High-throughput screening using patient-derived tumor xenografts to predict
799 clinical trial drug response. *Nat Med* **21**, 1318-1325 (2015).

800 29 Anurag, M. *et al.* Proteogenomic Markers of Chemotherapy Resistance and Response in
801 Triple-Negative Breast Cancer. *Cancer Discov* **12**, 2586-2605 (2022).

802 30 Krug, K. *et al.* Proteogenomic Landscape of Breast Cancer Tumorigenesis and Targeted
803 Therapy. *Cell* **183**, 1436-1456 e1431 (2020).

804 31 Ruprecht, B. *et al.* A mass spectrometry-based proteome map of drug action in lung cancer
805 cell lines. *Nat Chem Biol* **16**, 1111-1119 (2020).

806 32 Jumper, J. *et al.* Highly accurate protein structure prediction with AlphaFold. *Nature* **596**,

583-589 (2021).

33 Abramson, J. *et al.* Accurate structure prediction of biomolecular interactions with AlphaFold 3. *Nature* **630**, 493-500 (2024).

34 Theodoris, C. V. *et al.* Transfer learning enables predictions in network biology. *Nature* **618**, 616-624 (2023).

35 Cui, H. *et al.* scGPT: toward building a foundation model for single-cell multi-omics using generative AI. *Nat Methods* (2024).

36 Lamb, J. *et al.* The Connectivity Map: using gene-expression signatures to connect small molecules, genes, and disease. *Science* **313**, 1929-1935 (2006).

37 Subramanian, A. *et al.* A Next Generation Connectivity Map: L1000 Platform and the First 1,000,000 Profiles. *Cell* **171**, 1437-1452 e1417 (2017).

38 Sun, R. *et al.* Accelerated Protein Biomarker Discovery from FFPE Tissue Samples Using Single-Shot, Short Gradient Microflow SWATH MS. *J Proteome Res* **19**, 2732-2741 (2020).

39 Sun, R. *et al.* A prostate cancer tissue specific spectral library for targeted proteomic analysis. *Proteomics* **22**, e2100147 (2022).

40 Zhong, Q. *et al.* Proteomic-based stratification of intermediate-risk prostate cancer patients. *Life Sci Alliance* **7** (2024).

41 Sun, R. *et al.* Proteomic Dynamics of Breast Cancer Cell Lines Identifies Potential Therapeutic Protein Targets. *Mol Cell Proteomics* **22**, 100602 (2023).

42 Kanehisa, M. & Goto, S. KEGG: kyoto encyclopedia of genes and genomes. *Nucleic Acids Res* **28**, 27-30 (2000).

43 Zhou, Y. *et al.* Metascape provides a biologist-oriented resource for the analysis of systems-level datasets. *Nat Commun* **10**, 1523 (2019).

44 Szklarczyk, D. *et al.* The STRING database in 2023: protein-protein association networks and functional enrichment analyses for any sequenced genome of interest. *Nucleic Acids Res* **51**, D638-D646 (2023).

45 Kramer, A., Green, J., Pollard, J., Jr. & Tugendreich, S. Causal analysis approaches in Ingenuity Pathway Analysis. *Bioinformatics* **30**, 523-530 (2014).

46 Cao, Y., Charisi, A., Cheng, L. C., Jiang, T. & Girke, T. ChemmineR: a compound mining framework for R. *Bioinformatics* **24**, 1733-1734 (2008).

47 Lundberg, S. M. & Lee, S.-I. in *Neural Information Processing Systems*.

48 Bliss, C. I. The toxicity of poisons applied jointly 1. *Annals of applied biology* **26**, 585-615 (1939).

Figure 1

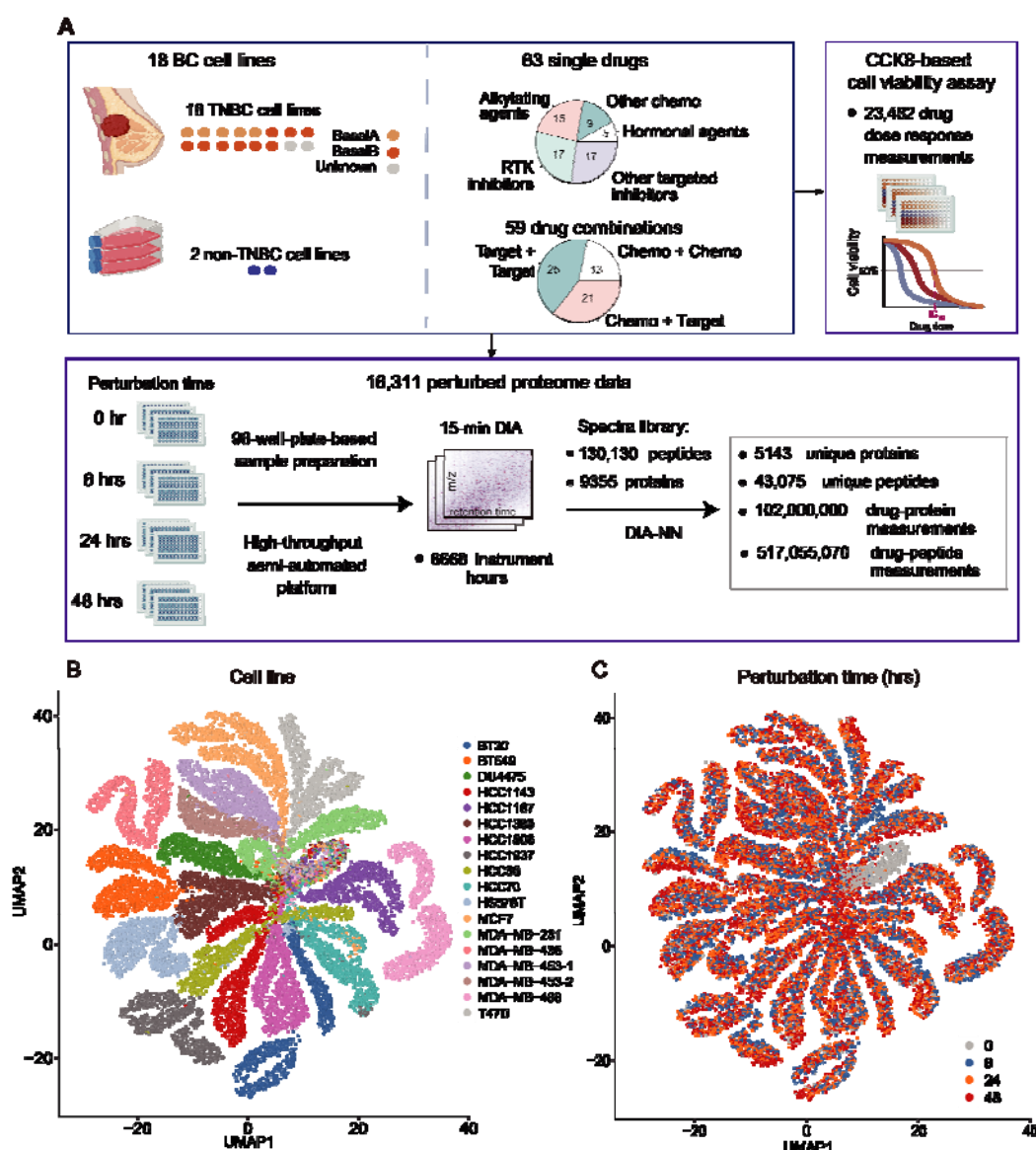
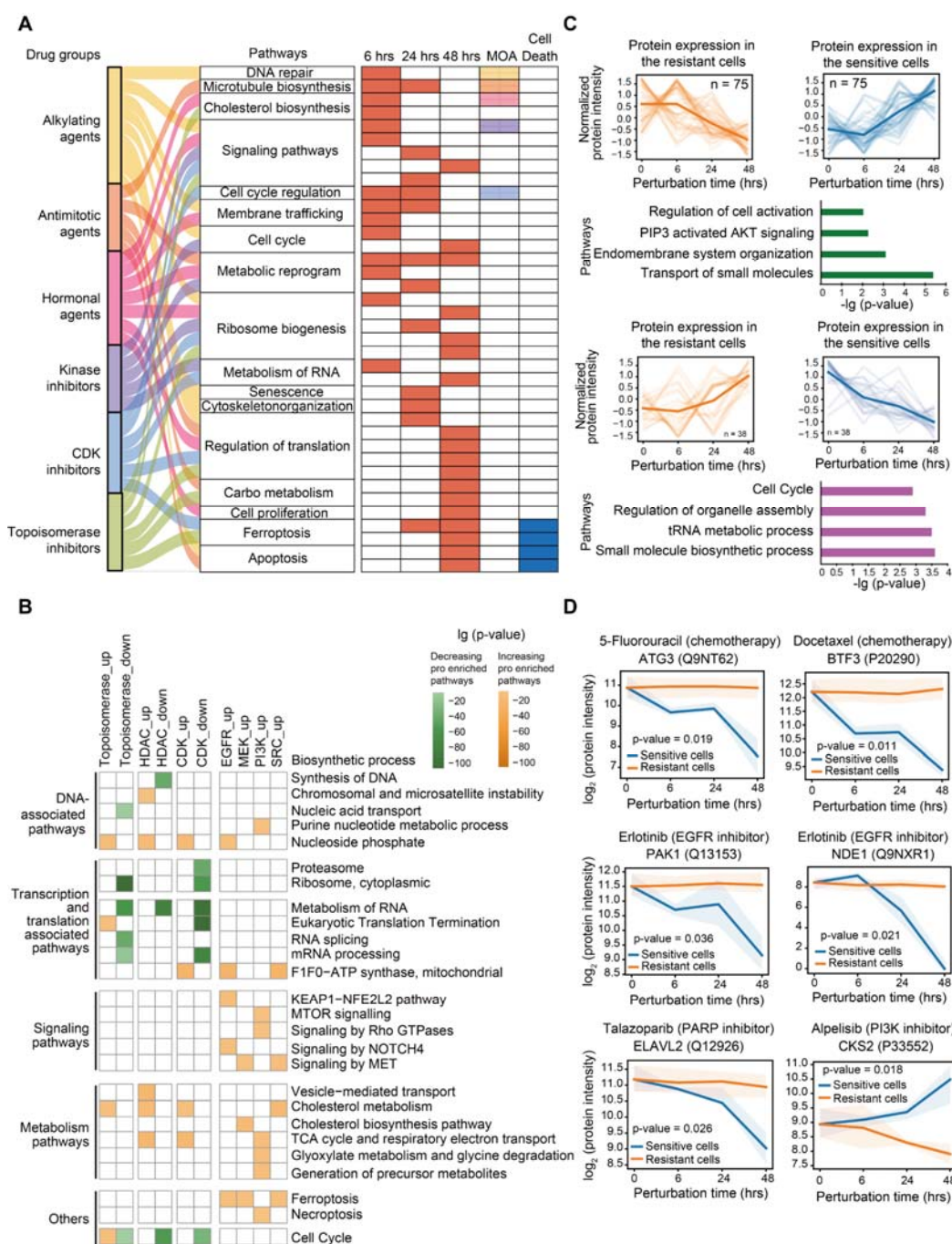


Figure 1. Overview of this perturbation proteomics study.

(A) Study Design: Our perturbation proteomics approach involved an extensive panel of 18 cell lines, including 16 triple-negative breast cancer (TNBC) cell lines and 2 non-TNBC cell lines, which were exposed to a battery of 63 FDA-approved drugs over various time spans. Each condition was biologically replicated thrice, yielding a dataset encompassing 16,311 proteomic profiles and 23,554 cell viability assessments. (B)-(C) UMAP shows the unsupervised clustering of the whole proteomes by different cell lines (B), perturbation durations (C).

853 **Figure 2**



854
855 **Figure 2. Longitudinal proteomic responses to drug perturbation.** (A) The top five
856 pathways enriched by differentially regulated proteins, as determined by PertScore (see
857 **Figure S2D**), over various treatment intervals. Pathways associated with the mechanism of
858 action (MOA) and cell death were specifically annotated. (B) Pathways enriched by
859 proteins that were consistently upregulated or downregulated post-drug perturbation.
860 Pathways associated with proteins exhibiting a decrease in abundance are represented in
861 green, while those associated with proteins with increased abundance are shown in orange.
862 (C) Clusters of proteins exhibiting sustained dysregulation throughout the duration of

863 chemotherapy drug exposure, with an emphasis on reversed trends between resistant and
 864 sensitive cell groups. The enriched pathways related to these protein clusters are also
 865 depicted. **(D)** Temporal changes in protein levels that were distinctively expressed between
 866 resistant and sensitive cell groups. The p-value indicates the significance of the interaction
 867 between time points and the response of sensitive versus resistant cells, as assessed by
 868 two-way ANOVA.

869

870

Figure 3

A

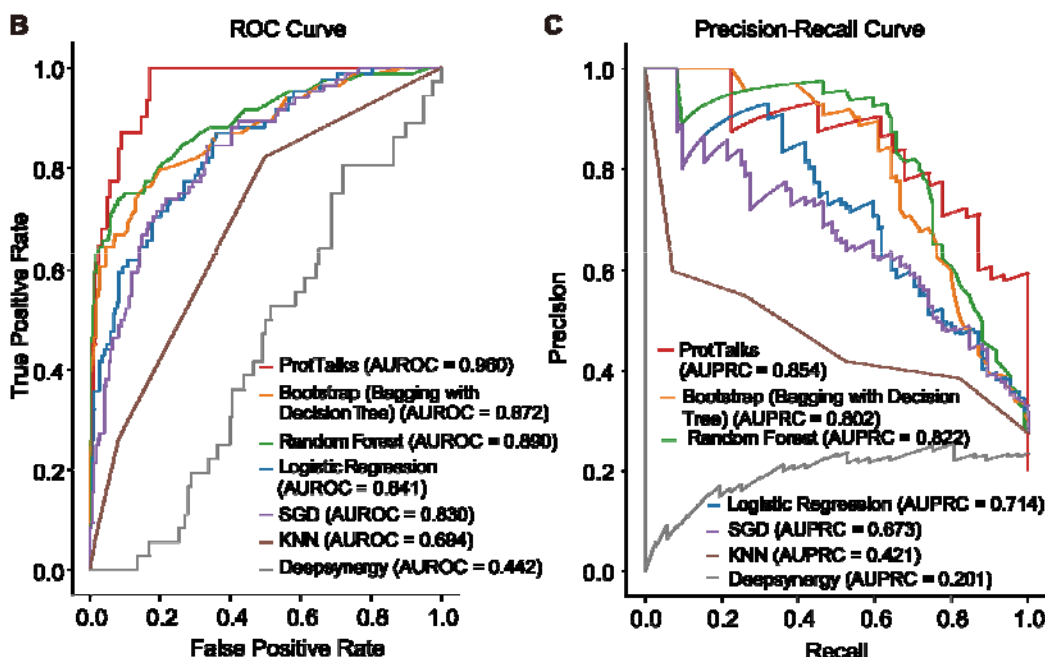
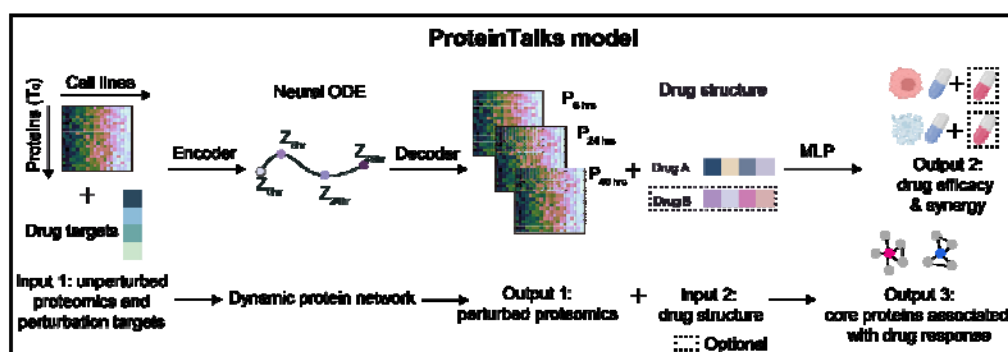
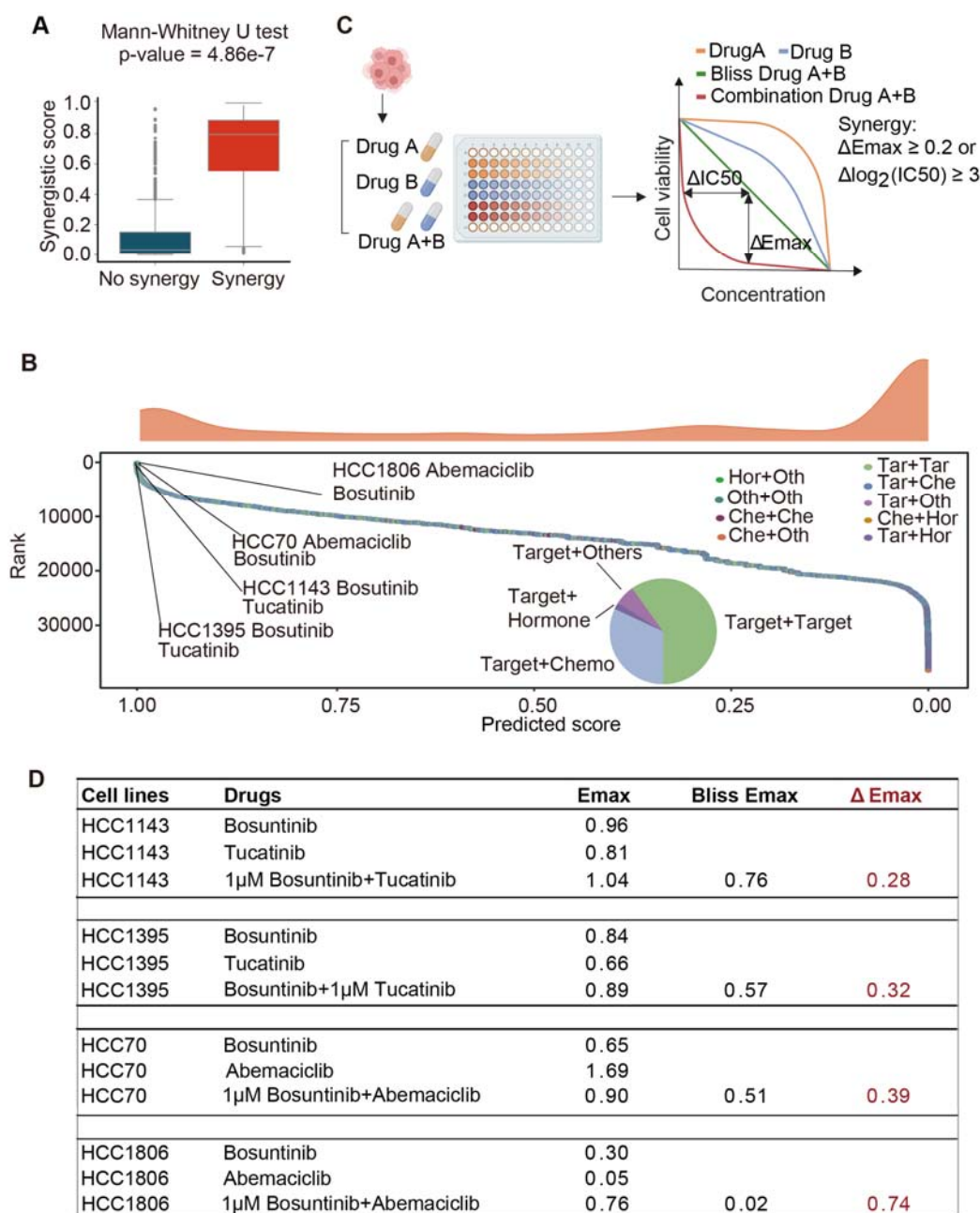


Figure 3. Development and performance of the ProteinTalks model. (A) The architecture of ProteinTalks. The baseline proteome from untreated cell lines and the list of drug targets are encoded into ProteinTalks, a linear network, to predict the perturbed proteomes, which are then decoded. Then the predicted proteome output from the first module, along with the initial time point proteome data and drug features obtained from SMILES descriptors, are processed through a linear layer to predict the essential proteins associated with drug response and effectiveness of single drugs or drug combinations. MLP, multilayer perception. (B)-(C) AUROC (B) and AUPRC (C) of six models with the ProteinTalks model.

884 **Figure 4**

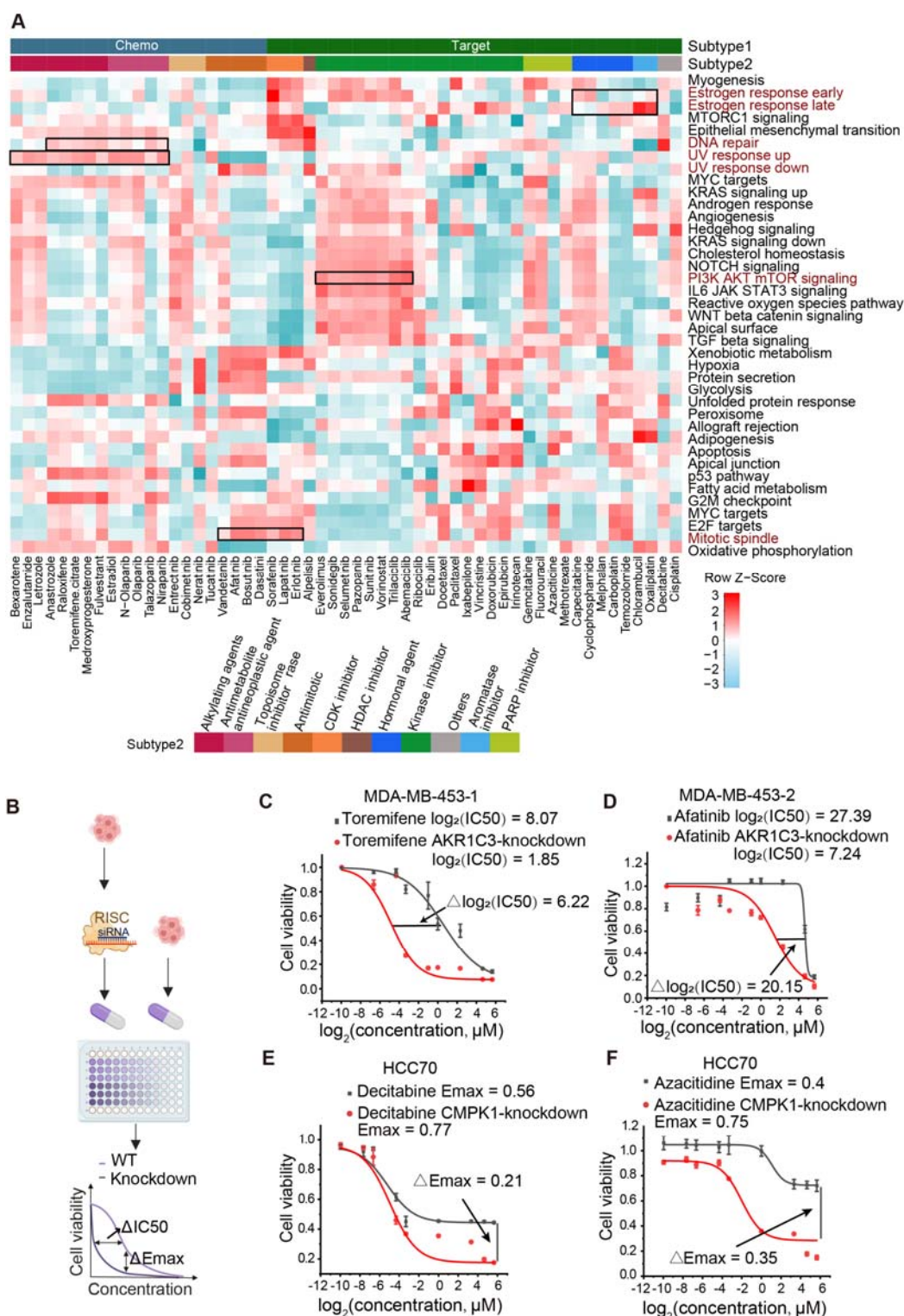


885

886 **Figure 4. Deciphering the ProteinTalks Model's Predictive Power for Drug**
887 **Combination Synergy.** (A) Deciphering the ProteinTalks' Predictive Power for Drug
888 Combination Synergy¹⁶. Statistical significance was assessed using the Mann-Whitney U
889 test. (B) Ranking of drug combinations based on their predicted synergy scores, with the
890 highly synergistic combinations explicitly highlighted. The distribution of drug groups
891 within the top 1000 synergistic combinations, as forecasted by the ProteinTalks model, is
892 depicted in a pie chart. Tar, target drug; Che, chemotherapy drug; Hor, hormonal drug; Oth,
893 other drug. (C) Description of the cytotoxicity assay protocol utilized for evaluating drug
894 combinations and the specific criteria applied to determine drug synergy. (D) Results from

895 cytotoxicity assays for drug combinations predicted by the ProteinTalks model to be
 896 synergistic, including combinations of bosutinib with tucatinib in HCC1395 and HCC1143
 897 cells, and bosutinib with abemaciclib in HCC70 and HCC1806 cells. The treatments were
 898 administered as follows: HCC1143 cells received a combination of bosutinib and tucatinib
 899 with a fixed concentration of 1 μ M bosutinib; HCC1395 cells were treated with the same
 900 drugs but with a fixed concentration of 1 μ M tucatinib; and HCC70 and HCC1806 cells
 901 were exposed to a combination of abemaciclib and bosutinib with a fixed concentration of 1
 902 μ M bosutinib. Each assay plate included triplicate wells. The shifts in efficacy (ΔE_{max}),
 903 representing the reduced cell viability, were determined by calculating the difference in
 904 efficacy between the observed combination response and the expected response based on
 905 Bliss independence⁴⁸.
 906

907 **Figure 5**



908

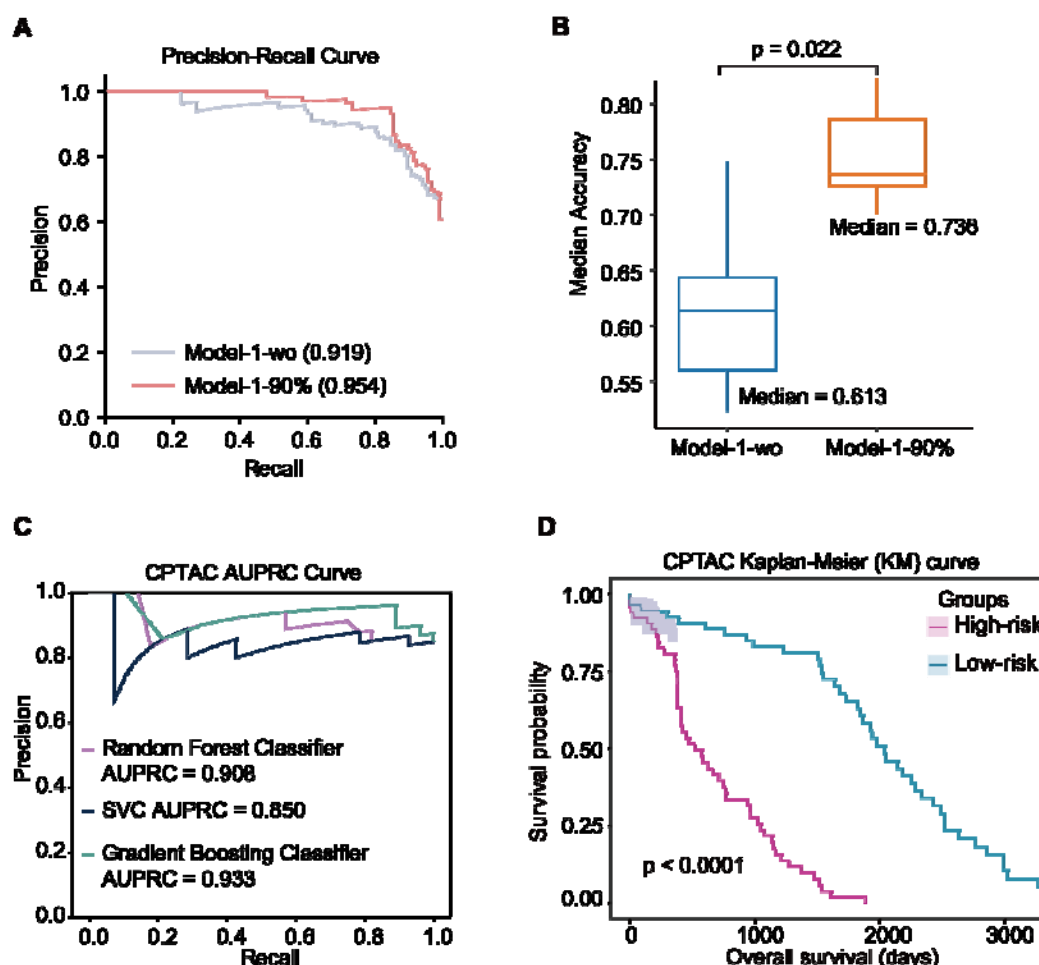
909 **Figure 5. Exploring the Interpretability of the ProteinTalks Model. (A)** Heatmap shows
910 the SHAP values of each pathway for each drug after z-score normalization. Each row
911 represents different protein pathways, and each column represents different drugs.

912 Hierarchical clustering was performed using Euclidean distance and the complete linkage
 913 agglomeration method. Different colors in the columns indicate different categories of
 914 drugs. **(B)** Schematic of the cytotoxicity assay procedure, which incorporates
 915 siRNA-mediated gene knockdown in cell lines. **(C)** Outcomes of cytotoxicity assays on two
 916 MDA-MB-453 cell lines and HCC70 cells, post-transfection with AKR1C3 and CMPK1
 917 siRNAs, followed by treatment with toremifene, afatinib, azacytidine, and decitabine,
 918 respectively.

919

920

Figure 6

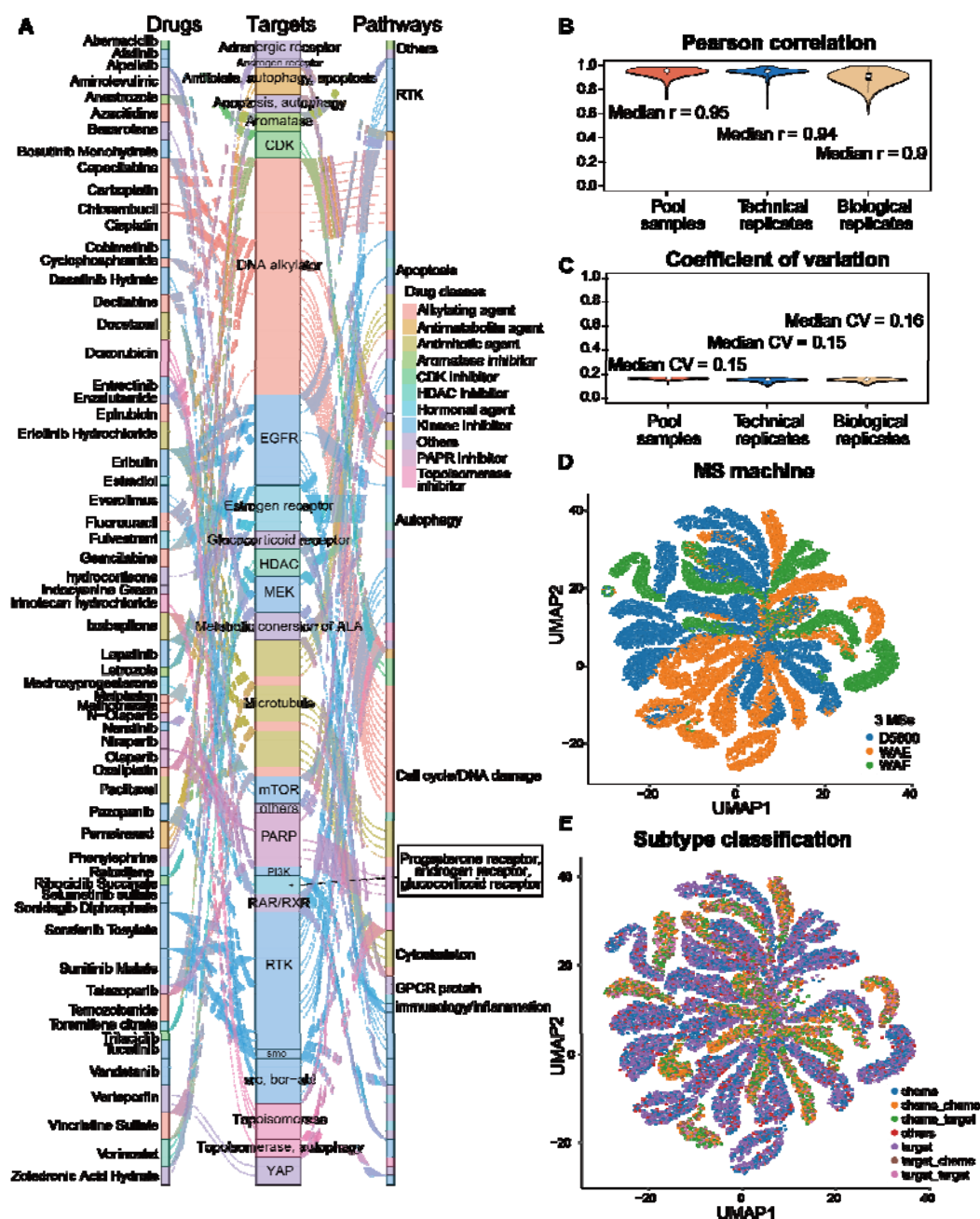


922

923 **Figure 6. Assessing the Clinical Relevance of the Model-1.** (A) In the BC patient-derived
 924 tumor cells (PDTCs)' transcriptomic dataset, the AUPRC of the ProteinTalks model's drug
 925 efficacy prediction performance between model-1-wo and model-1-90%. model-1-wo,
 926 model-1-without, was trained from scratch solely on the same subset of transcriptomic PDTc
 927 data without transfer learning from the perturbation proteomic data (Figure S8A). When 90%
 928 of the PTDS perturbation proteomic data was used in the first stage, the corresponding
 929 models obtained was model-1-90% (Figure S8A). (B) In the pan-cancer PDTXs'
 930 transcriptomic dataset, the accuracy of the model's drug efficacy prediction performance
 931 between model-1-wo and model-1-90%. model-1-wo, model-1-without, was trained from
 932 scratch solely on the same subset of transcriptomic PDTc data without transfer learning from
 933 the perturbation proteomic data (Figure S8A). When 90% of the PTDS perturbation
 934 proteomic data was used in the first stage, the corresponding models obtained were
 935 model-1-90% (Figure S8A). Statistical significance was determined via paired Welch's
 936 t-test. (C) In the CPTAC proteomic dataset (N=107), the AUPRCs for the prognosis
 937 prediction performance in the test dataset (N=33) were evaluated using different machine
 938 learning models based on top the 60 proteins identified by ProteinTalks. (D) In the CPTAC
 939 proteomic dataset, the Kaplan-Meier (KM) curves of overall survival based on the top 60

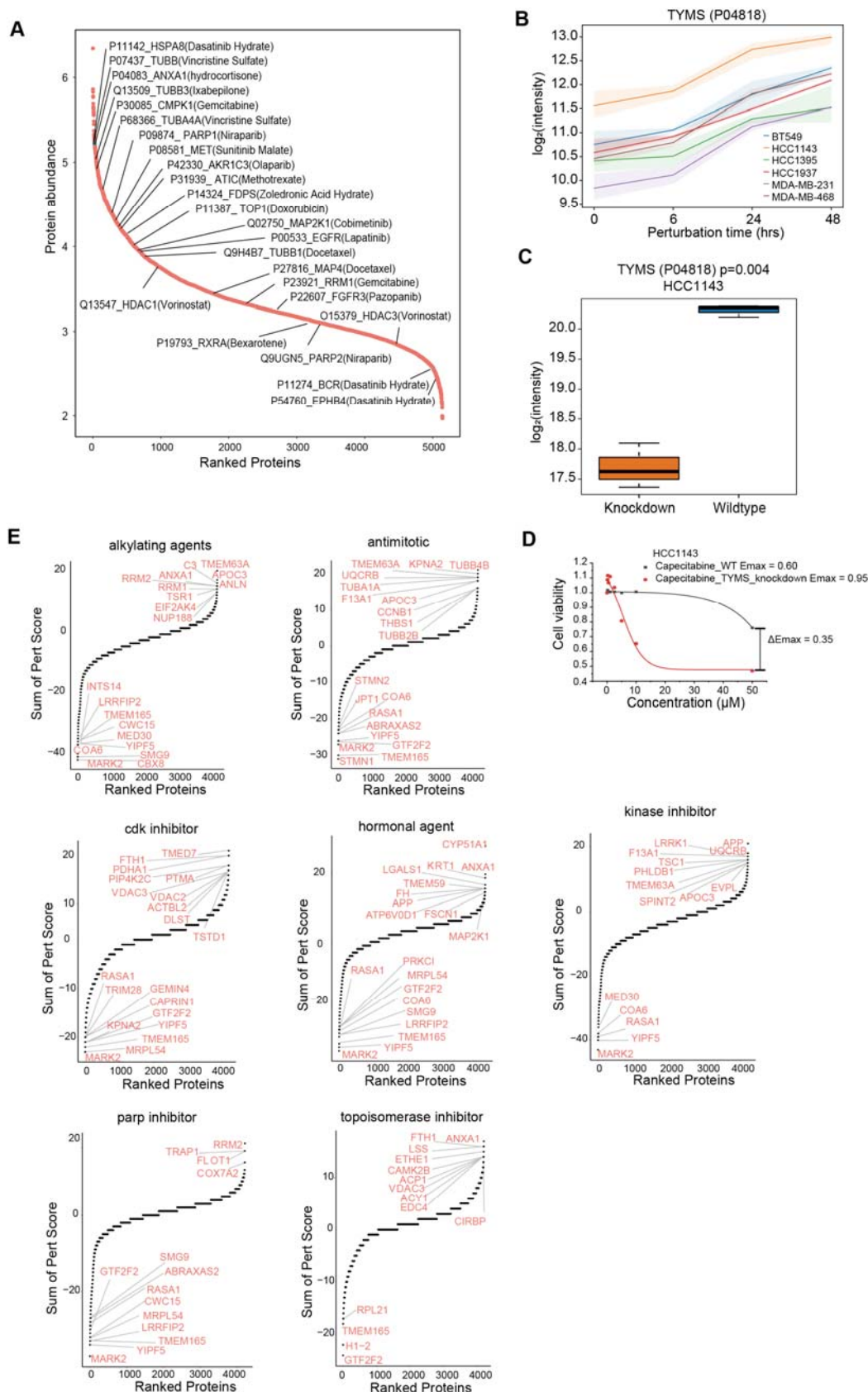
940 proteins identified by ProteinTalks.

Supplementary Figure 1



Supplementary Figure 1. Quality control analysis (A) Drug classification for 63 drugs. (B)-(C) The reproducibility of pool, technical replicate, and biological samples evaluated by Pearson correlation (B) and coefficient of variation (C). (D)-(E) UMAP shows whole proteomics grouping by MS machine (D) and drug classification (E).

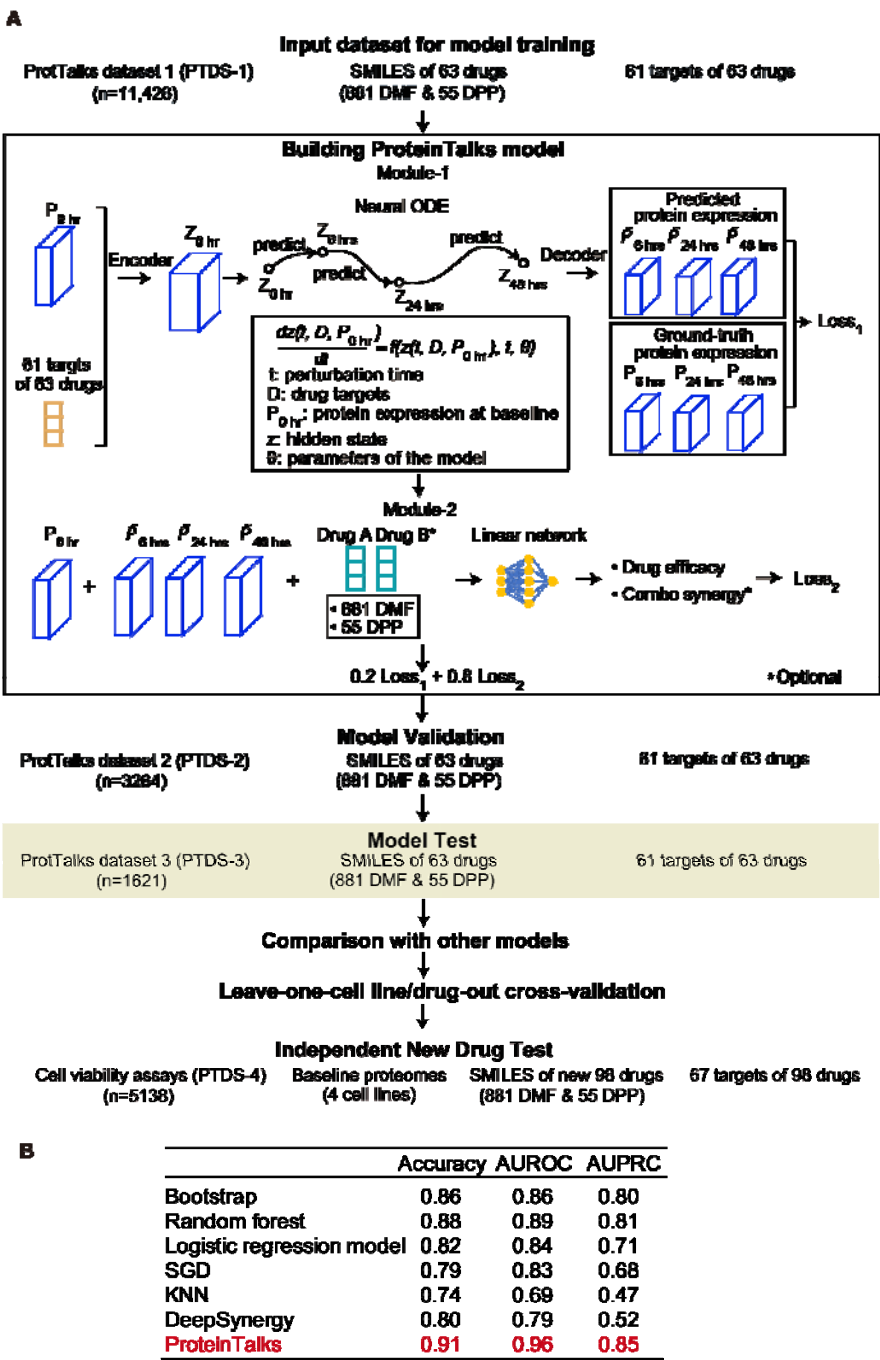
949 Supplementary Figure 2



950

951 **Supplementary Figure 2. Dysregulated perturbation score (A)** The targeted proteins are
 952 identified in this perturbation proteomics dataset. **(B)** TYMS protein expression after
 953 treatment. **(C)** Box plots depicting the log2-scaled protein abundance detected by MS to
 954 evaluate RNA interference efficacy. The knockdown efficiency of TYMS was evaluated in
 955 HCC1143 cells. **(D)** Cytotoxicity assay results of HCC1143 cells interfered by TYMS
 956 siRNA then treated with capecitabine. **(E)** Perturbation score for different drug classes.

957 **Supplementary Figure 3**

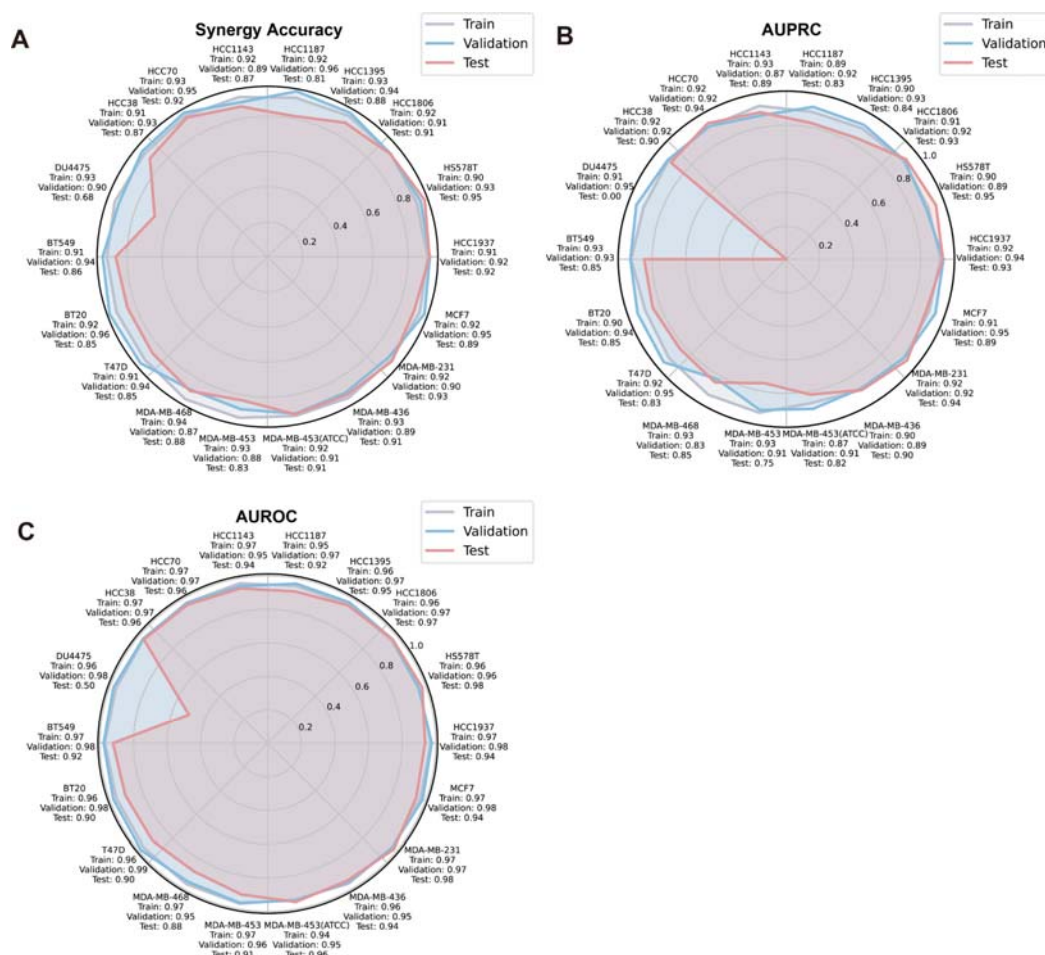


958

959 **Supplementary Figure 3. Detailed workflow of ProteinTalks building.** Schematic
960 representation of the process involved in constructing the ProteinTalks model. The 16,311
961 perturbation proteomic data were randomly divided into a training dataset (PTDS-1,
962 n=11,426), a validation dataset (PTDS-2, n=3264), and a test dataset (PTDS-3, n=1621),
963 following a 7 : 2 : 1 split. After data preprocessing, PTDS-1 was used for training the model
964 through two modules, while PTDS-2 as validation set was used to optimize the model
965 parameters. The performance of the ProteinTalks model was evaluated using PTDS-3 and

966 compared with other models (**Figure 3B**). Leave-one-cell-line-out was used for
 967 cross-cell-line-validation (**Figure S4**), while leave-one-drug-out was used for
 968 cross-drug-validation (**Figure S5A-C**). Finally, a set of 98 new drugs was independently
 969 assayed for cell viability in four TNBC cell lines. This set was used to evaluate the
 970 predictive efficacy of the ProteinTalks model for new drugs (**Figure S5D**). DMF,
 971 dimensional drug molecular fingerprints; DPP, dimensional drug physicochemical properties.

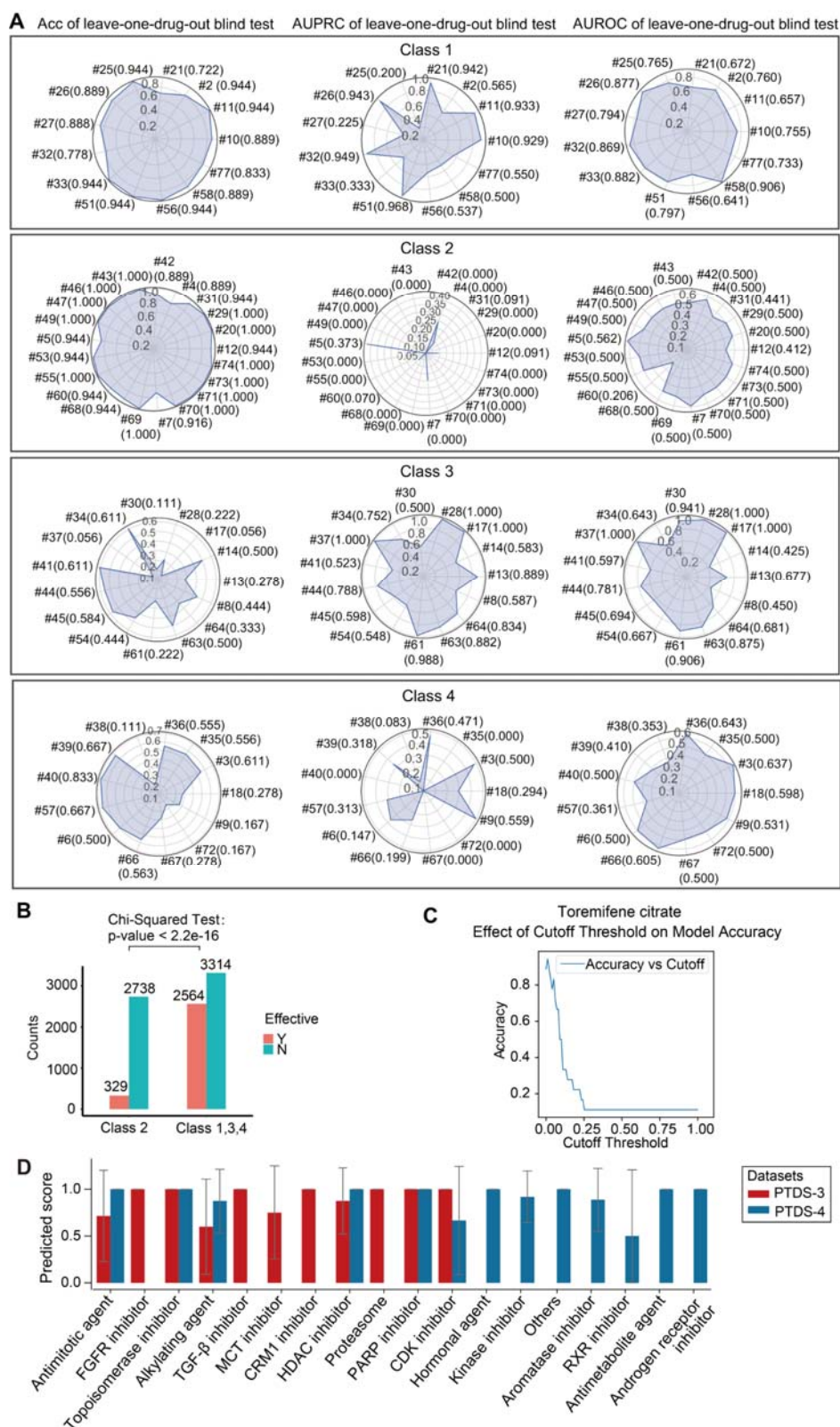
Supplementary Figure 4



973

974 **Supplementary Figure 4. Leave-one-cell line-out cross-validation of ProteinTalks.**
975 **(A)-(C)** Accuracy **(A)**, AUPRC **(B)**, and AUROC **(C)**. The radial plots display the
976 performance metrics for each cell line, including Synergy Accuracy, AUPRC (Area Under
977 the Precision-Recall Curve), and AUROC (Area Under the Receiver Operating
978 Characteristic Curve). Each plot shows the results for the training, validation, and test sets,
979 indicated by gray, blue, and red lines, respectively. Performance metrics are presented for
980 each cell line, indicating the model's ability to generalize across different contexts and
981 datasets. These metrics are crucial for assessing the predictive power and robustness of the
982 ProteinTalks model in protein interaction and drug efficacy studies.

Supplementary Figure 5

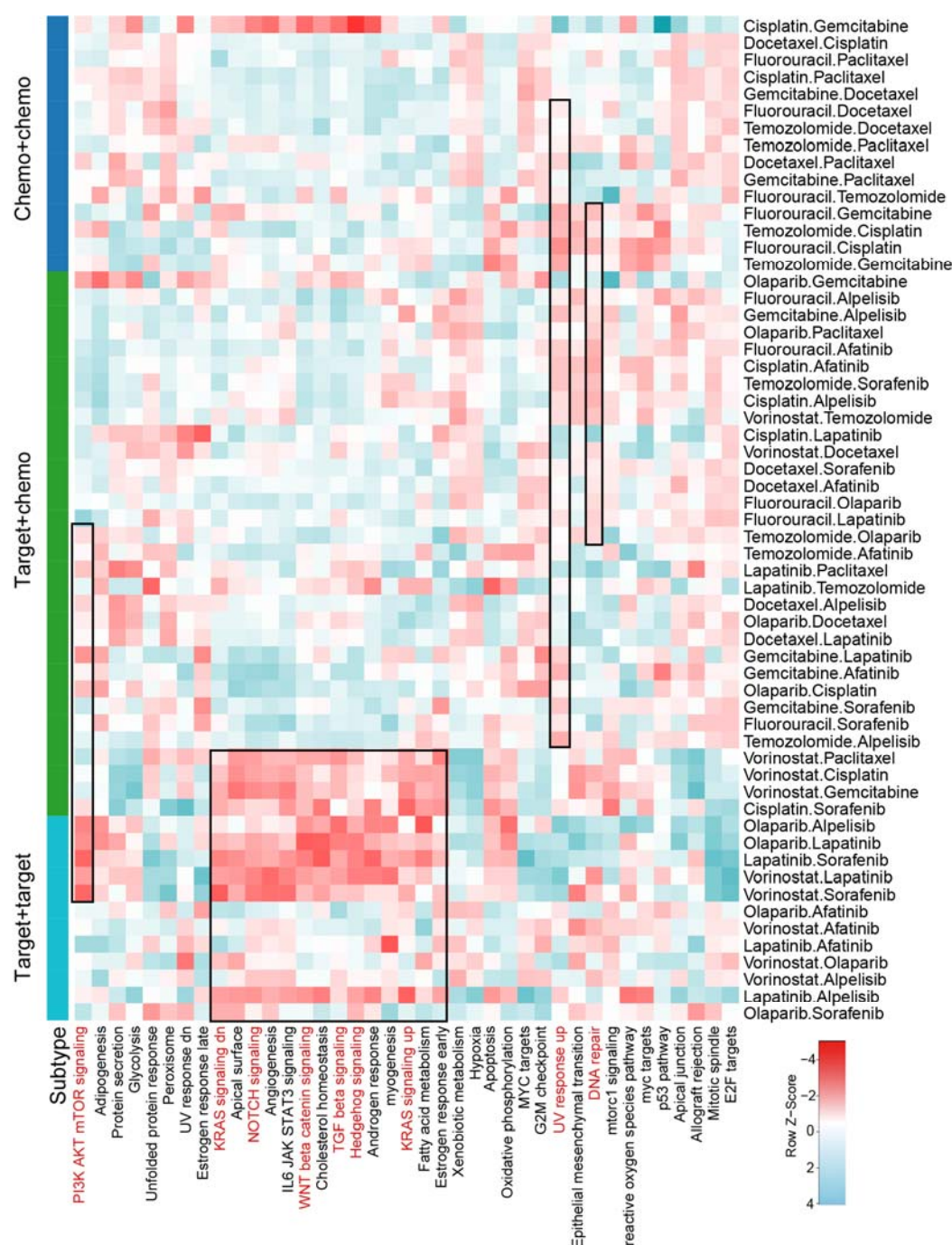


984

985 **Supplementary Figure 5. Testing of new drugs using ProteinTalks. (A)** The four classes

986 of drugs are defined by their accuracy (Acc), AUPRC, AUROC, which are determined
987 through leave-one-drug-out cross-validation. The radial plots display the performance
988 metrics for each cell line, including Synergy Accuracy, AUPRC (Area Under the
989 Precision-Recall Curve), AUROC (Area Under the Receiver Operating Characteristic
990 Curve), and Proteomics Multi-time Correlation. Each plot shows the results for the test
991 datasets and the median value are shown in the brackets. Performance metrics are presented
992 for each cell line, indicating the model's ability to generalize across different contexts and
993 datasets. The metrics are crucial for assessing the predictive power and robustness of the
994 ProteinTalks model in protein interaction and drug efficacy studies. **(B)** The curve shows the
995 effect of the cutoff threshold on the accuracy of the ProteinTalks model for predicting the
996 efficacy of toremifene citrate. **(C)** Columns display the counts of effective drugs and
997 ineffective drugs for each cell line in class 2 alone, as well as in combination with class 1, 3,
998 4. The Chi-Squared test was used to compare the differences between the effective and
999 ineffective drugs for each cell line in two groups: class 2 alone, and the combination of
1000 class 1,3,4 (p-value < 2.2e-16). **(D)** Comparative display of prediction scores for each drug
1001 group generated by the ProteinTalks model for PDTS-3 and PDTS-4.

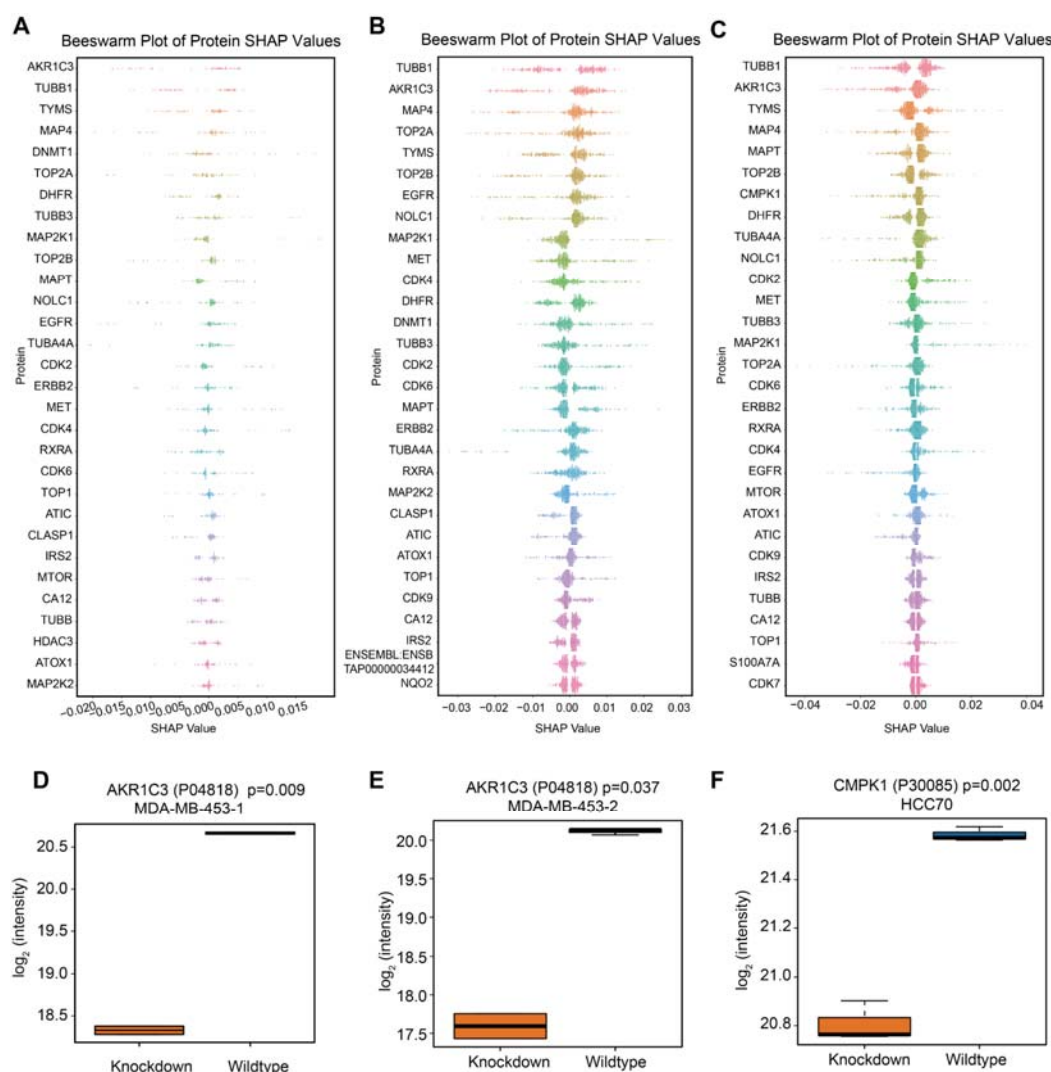
Supplementary Figure 6



Supplementary Figure 6. Pathway SHAP values for drug combinations. This heatmap shows the normalized pathway SHAP values, which is calculated using ProteinTalks for drug combinations after the training process. Heatmap shows the SHAP values of each pathway for each drug combination after z-score normalization. Each row represents different protein pathways, and each column represents different drug combinations. Hierarchical clustering was performed using Euclidean distance and the complete linkage agglomeration method. Different colors in the columns indicate different categories of drug

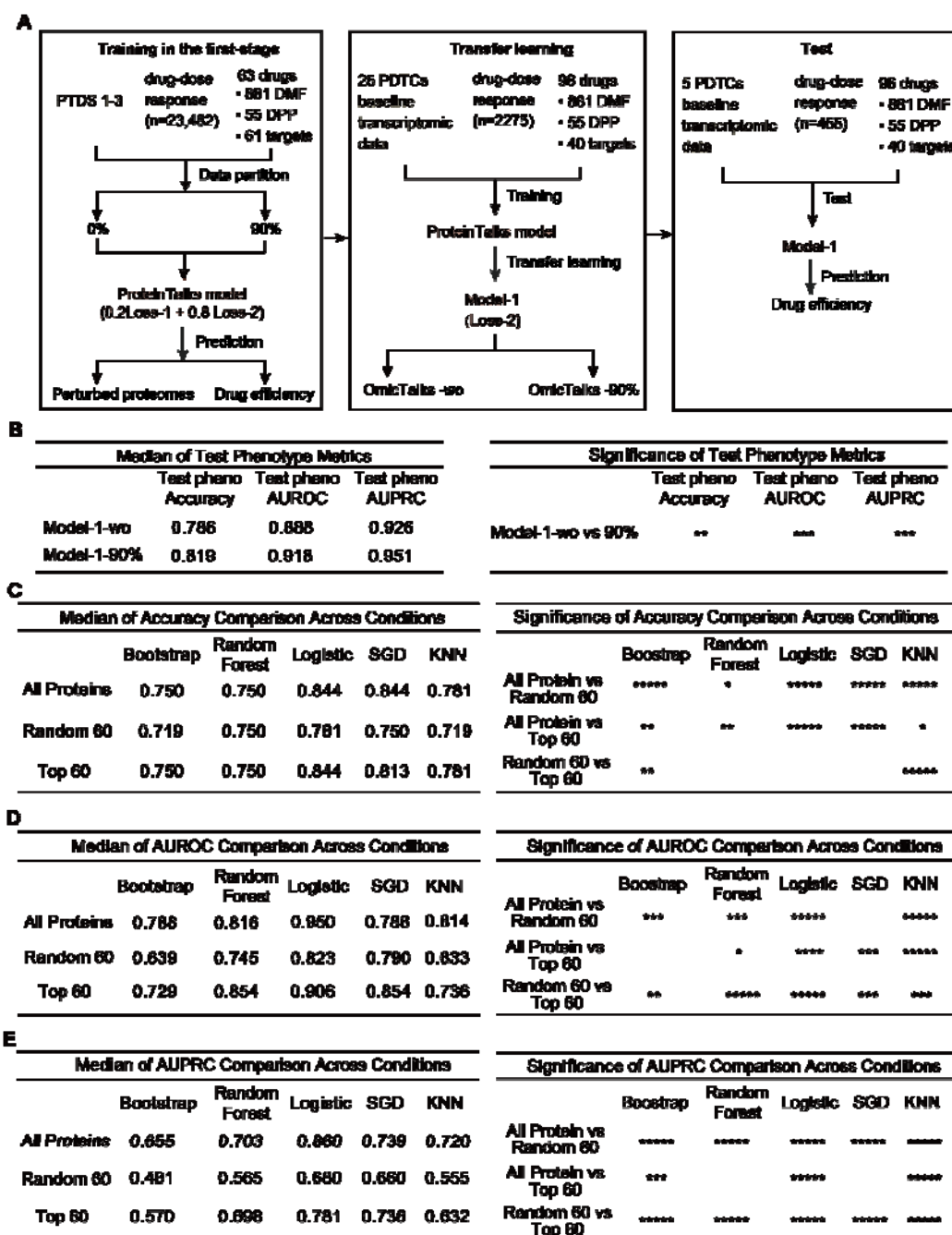
1011 combinations.

Supplementary Figure 7



Supplementary Figure 7 Interpretation of ProteinTalks model using SHAP values for different drugs. (A)-(C) Beeswarm plots illustrate the protein-level contributions to the model's predictions for various drugs, including hormonal agents (A), kinase inhibitors (B), and alkylating agents (C). Each point represents an individual protein's contribution, with different colors indicating different proteins. The x-axis represents the SHAP value, while the y-axis displays the corresponding proteins. Positive SHAP values signify a positive contribution to the predicted efficacy, whereas negative values indicate a negative contribution. The absolute value of the SHAP score represents the magnitude of the contribution, with larger absolute values indicating a more significant influence on the predicted efficacy. (D)-(F) Box plots depicting the log₂-scaled protein abundance detected by MS to evaluate RNA interference efficacy. The knockdown efficiency of AKR1C3 was evaluated in MDA-MB-453-1 cells (D), MDA-MB-453-2 cells (E), and CMPK1 in HCC70 cells (F).

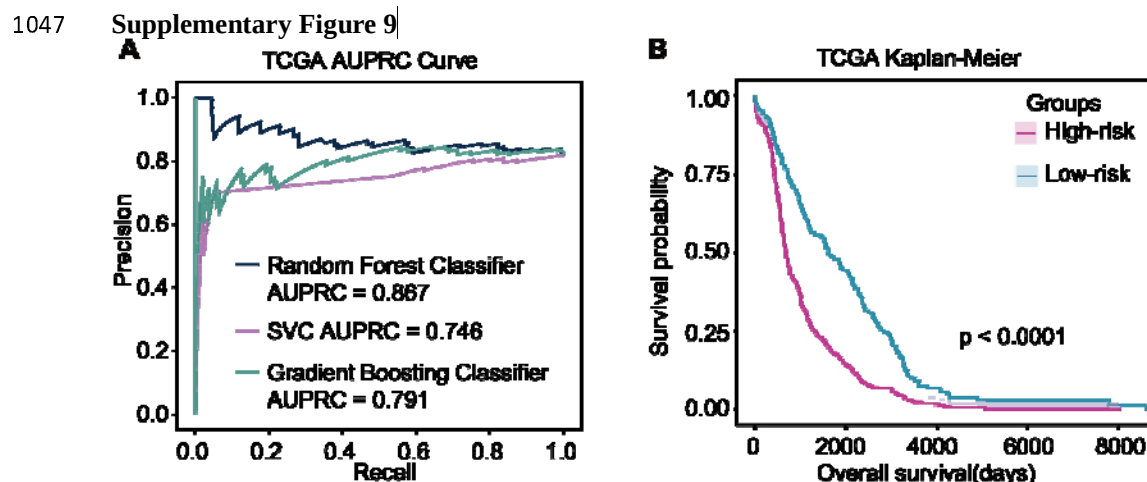
1028 **Supplementary Figure 8**



1029

1030 **Supplementary Figure 8. Assessing the clinical relevance of the model-1 with PDX**
 1031 **transcriptomics data. (A)** Diagram illustrating the construction of the ProteinTalks model.
 1032 The model underwent training with perturbation proteomic data and was subsequently
 1033 refined using a subset of transcriptomic data from PDX models. The remaining
 1034 transcriptomic data were used to evaluate the model's capability to predict drug efficacy.
 1035 Model-1-wo, Model-1-without, was trained from scratch solely on the same subset of
 1036 transcriptomic PDTc data without transfer learning from the perturbation proteomic data.
 1037 When 90% of the PTDS-1, -2, and -3 perturbation proteomic data were used in the first

1038 stage, the corresponding models obtained were Model-1-wo and Model-1-90%, respectively.
 1039 **(B)** Evaluation of the model's drug efficacy prediction performance, with pretraining
 1040 conducted using varying percentages of perturbation proteomic data. Statistical significance
 1041 was determined via the Mann-Whitney U test. **(C)-(E)** Performance comparison of drug
 1042 combination synergy predictions in TNBC patient cohorts, utilizing the complete proteome,
 1043 a random selection of 60 proteins, and the top 60 proteins identified by SHAP values within
 1044 the ProteinTalks. Significance levels were assessed using the Mann-Whitney U test, with
 1045 p-values denoted as follows: *, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$; ****, $p < 0.0001$.
 1046



1048

1049 **Supplementary Figure 9. Clinical relevance validation in TCGA dataset. (A)** In the
 1050 TCGA transcriptomic dataset (N=823), the AUPRCs for the prognosis prediction
 1051 performance in the test dataset (N=165) were evaluated using different machine learning
 1052 models based on the top 60 proteins identified by ProteinTalks. **(B)** In the TCGA proteomic
 1053 dataset, the Kaplan-Meier (KM) curves of overall survival based on the top 60 proteins
 1054 identified by ProteinTalks.